Intrahousehold Inequality and Savings in Retirement*

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Abstract

The distribution of resources between spouses and the extent to which household savings reflect each spouse's motives are fundamental to analyzing married retirees' welfare. In this paper, we estimate a dynamic collective model of retired couples with individual-specific preferences and relative bargaining power. Spouses jointly choose consumption and saving in the face of health and mortality shocks, with housing modeled as a distinct asset. We use the estimated model to provide one of the first quantitative assessments of intrahousehold inequality in bargaining power and its implications for lifetime consumption, savings, and welfare in old age. We find that wives' mean lifetime consumption shares are close to parity, although there is substantial dispersion. Within couples, wives have stronger precautionary motives and therefore greater incentives to save on average. As a result, policies that encourage faster dissaving of retirement wealth disadvantage wives relative to husbands, revealing distributional consequences that would be obscured under a unitary framework.

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1 Introduction

Individual welfare in retirement depends critically on access to adequate consumption and savings. For married couples, these choices are made jointly, so the balance of power within the household plays a central role in shaping each spouse's well-being. How resources are allocated between spouses, and the extent to which household savings reflect each spouse's motives, are fundamental to analyzing retirees' welfare. Old-age support policies can also generate differential welfare effects depending on each spouse's preferences and risks. For instance, means-tested programs that create spend-down incentives may disproportionately disadvantage the spouse with a stronger desire to save. Yet when outcomes are evaluated only at the household level, these distributional consequences are obscured.

Addressing questions about intrahousehold inequality and individual welfare requires moving beyond a unitary framework, which endows the married couple with a single household utility function and masks gender differences in preferences and power.[1] This paper develops and estimates a dynamic collective framework for studying couples' decisions in retirement, contributing one of the first quantitative assessments of intrahousehold inequality in bargaining power and its implications for lifetime consumption, savings, and welfare in old age. We specify each spouse's preferences and assume that decisions are made by maximizing a weighted sum of these values. The key input to this process is each spouse's Pareto weight, which reflects relative bargaining power and is shaped by features of the environment such as relative income shares. The collective framework allows us to examine how relative bargaining power varies across households, insights that unitary models cannot deliver. We identify variation in Pareto weights using income shares as a distribution factor, a variable that influences the bargaining process but not preferences or the budget set. To help identify the weights, we also exploit shifts in household structure, such as a spouse's death or nursing home entry.

We embed this collective framework in a dynamic model that explicitly incorporates marital transitions and housing. With mortality risk, married individuals transition into singlehood after the death of a spouse. It is therefore necessary to incorporate singles' behavior when solving the intertemporal problem for married households. Crucially, we exploit variation by gender among single retirees to help identify gender-specific preferences. While couples' decisions reflect both preferences and relative bargaining power, singles' choices reveal preferences more directly as they are not confounded by intrahousehold bargaining. We therefore use singles' behavior to help identify gender-based heterogeneity in preferences.

[1] Throughout the paper, our analysis of couples focuses on married different-sex couples.

The model also treats housing as a distinct asset in retirees' savings decisions. Although a large share of retirement wealth is held in housing, much of the existing literature treats all assets as liquid. Accounting for the unique features of housing is especially important when analyzing heterogeneous savings motives between spouses. Because housing assets are harder to dissave than liquid assets, the value of homeownership may vary between spouses depending on their relative savings motives. Housing also plays a central role in old-age welfare programs that exempt home equity from means tests, such as Medicaid. The welfare implications of these policies may differ across spouses, depending on who benefits relatively more from the exemption. By modeling housing explicitly, we capture how asset composition interacts with intrahousehold dynamics and policy design in shaping individual welfare.

Using Health and Retirement Study (HRS) data covering 1998–2014, we begin by examining whether married retirees' consumption–savings behaviors are consistent with a collective framework rather than a unitary model. Under the unitary model, conditional on total resources, household choices should not vary with distribution factors such as individual income shares. In contrast to this prediction, we find that the wife's income share is negatively correlated with household consumption and positively correlated with household savings. These patterns are inconsistent with the unitary model but are consistent with a collective framework in which the wife's bargaining power rises with her relative income share. This interpretation reflects wives' stronger precautionary savings motives, driven by their longer life expectancy and greater likelihood of needing long-term care at a time when no spouse is available to provide it.

We next document a series of descriptive patterns that guide the modeling choices for our collective framework, which explicitly incorporates marital transitions and housing. Couples and singles differ sharply in how they hold and decumulate housing assets: homeownership remains high among couples well into advanced ages, while singles rapidly dissave housing. We document lack of spousal care and higher nursing home risk for singles as a potential reason for their lower homeownership. Institutional rules amplify these differences by marital status. Medicaid, a means-tested public insurance program that covers long-term care costs, exempts home equity from the means test for married nursing home residents but allows cost recovery from the housing assets of single beneficiaries, a feature that discourages homeownership among singles relative to couples (Greenhalgh-Stanley, 2012). We also show gender differences in consumption-savings behaviors among single retirees: single women in lower income groups consume less than observably similar men, while single women in higher income groups save less than their male counterparts.

Building on the descriptive findings, we develop a dynamic model of retirees' savings, housing, and long-term care decisions under health and mortality risk. The model accommodates different initial household types, including couples, single women, and single men. For married households, we use a collective framework in which husbands and wives have separate preferences. The model allows for gender heterogeneity not only in health and mortality risk, but also in preferences for caregiving, consumption-housing tradeoff, and bequest motives toward children. Couples' choices maximize a weighted sum of spouses' values, with Pareto weights that depend on the wife's income share and household permanent income (PI), allowing intrahousehold bargaining power to vary across households. Because spouses have distinct preferences, the model delivers separate value functions for husbands and wives, which we use to assess policies from an individual rather than purely household perspective. Means-tested social insurance programs, representing Medicaid, are incorporated and treat housing differently by marital status: single nursing home residents must effectively use their housing wealth to finance care, whereas married households benefit from a homestead exemption.

We estimate the model using indirect inference, selecting auxiliary models that capture the data features most informative for identifying the structural parameters. To identify the Pareto weights, we exploit variation in the wife's income share. Because income shares affect choices only through the bargaining channel, differences in choices across married households with different shares identify how the weights respond to the distribution factor. To recover how bargaining power varies with PI, we exploit how the effect of income share on consumption changes across the PI distribution. To help identify the level of the Pareto weights, we exploit changes in household behavior around spousal nursing home entry, which increases the community spouse's influence over household decision making, and the resulting adjustment in household behavior provides information about the pre-entry distribution of bargaining power.

To help identify gender-specific preferences, we exploit variation by gender among single retirees, whose decisions are free from intrahousehold bargaining and therefore provide cleaner information about preference heterogeneity by gender. We use gender differences in housing choices among singles to discipline gender-specific consumption–housing tradeoffs and wealth trajectories of single men and women to inform gender-specific bequest motives toward children. In addition, spousal care rates provide particularly useful information for identifying caregiving utility and the late-life savings behavior of married couples provides key information for identifying altruism toward a surviving spouse.

The estimation results reveal substantial cross-sectional heterogeneity in wives' Pareto weights, with their bargaining position increasing with their income share and decreasing with PI. Regarding gender differences in saving motives, we find that women have stronger precautionary motives due to longer life expectancy and greater disability risk, while men have stronger bequest motives toward children when dying as a single retiree.

Using the estimated model, we first run simulations to quantify intrahousehold inequality in bargaining power and lifetime consumption over the marriage in retirement. We consider two cohorts separately: couples who entered retirement in 1998–2000 and those who entered retirement in 2012–2014. In the earlier cohort, women's mean relative Pareto weight is 0.446, while in the later cohort it is 0.477. In both cohorts, there is substantial dispersion, with the standard deviation roughly one-half of the corresponding mean. For each cohort, the distribution of wives' lifetime consumption shares closely mirrors the distribution of their Pareto weights, although dispersion in consumption shares is attenuated by consumption floors. Wives' mean consumption shares are 0.482 in the earlier cohort and 0.495 in the later cohort, suggesting that their mean shares over the lifetime of the marriage are close to parity.

Next, to examine heterogeneity in savings motives between spouses and how these motives translate into couples' observed savings, we conduct counterfactuals in which either the husband or the wife holds full bargaining power, so that household decisions reflect the preferences of one spouse alone. Under either scenario with extreme Pareto weights, the spouse whose preferences are fully weighted does not provide spousal care to avoid caregiving disutility, increasing the household's exposure to formal long-term care risk. As a result, household savings rise due to stronger precautionary motives, with the magnitude varying across PI groups (defined by PI terciles): means-tested public insurance dampens the savings response for low- and mid-PI households, while high-PI households experience much larger increases. Which counterfactual scenario leads to a larger increase in savings also differs by PI group. Among low- and mid-PI households, for whom bequest motives are less relevant and precautionary motives dominate, savings rise more when the wife holds full bargaining power. In contrast, among high-PI households, savings increase more when the husband holds full bargaining power, driven by his stronger bequest motives toward children.

Finally, we study reforms to Medicaid housing rules, focusing on gendered welfare effects within married households. We find substantial heterogeneity, with particularly strong effects for initial singles, for women, and for low-PI retirees. Importantly, our collective approach reveals within-household differences, with wives typically experiencing larger welfare changes. Among households for whom means-tested Medicaid is relevant, saving behavior is driven primarily by precautionary motives rather than bequest motives. Since housing assets decumulate slowly and wives have stronger precautionary savings motives, Medicaid's housing rules tend to affect them more. These gender disparities would be invisible under a unitary approach, underscoring the value of a collective model for evaluating policy reforms.

This paper contributes to a large empirical literature on savings in retirement.[2] Prior work has made substantial progress in identifying why retirees save, emphasizing precautionary motives arising from medical expense risk and bequest motives.[3] However, nearly all of this literature focuses on single retirees. Only a small set of studies examine couples' savings (e.g., Braun et al. (2017), Nakajima and Telyukova (2020), De Nardi et al. (2025)), and they typically rely on unitary frameworks that are based on aggregate household preferences. While existing work has studied differences in savings motives across household types and across the PI distribution, unitary models cannot quantify within-household heterogeneity in savings motives between spouses. Our contribution is to expand the role of gender and incorporate intrahousehold heterogeneity into the analysis of retirees' savings. By estimating a collective model, we show how savings motives differ within married households and how these differences are aggregated into household-level decisions through their respective Pareto weights.

Our analysis also connects to the literature on collective models, which began with the foundational work of Chiappori (1988), Chiappori (1992) and Apps and Rees (1988). In contrast to unitary frameworks in which the household is characterized by a unique utility function, collective approaches specify individual preferences and aggregate them using relative bargaining weights that place allocations on the Pareto frontier. Cross-sectional variation in Pareto weights is captured by their dependence on the economic environment, such as wages and non-labor income. The presence of a distribution factor (Browning and Chiappori (1998), Chiappori et al. (2002), Blundell et al. (2005)), such as income shares or sex ratios, facilitates identification because it affects only the Pareto weights and not preferences or the budget set. A large empirical literature exploits such cross-sectional variation to estimate collective models of the working-age population.[4] There is relatively little work on intrahousehold inequality among retirees, and the existing evidence is largely based on static frameworks; for example, Cherchye et al. (2012) use Dutch consumption survey data to estimate the consumption sharing rule among retired couples. We contribute to this literature by estimating a dynamic collective model of retirees and using it to quantify the implications of intrahousehold inequality for lifetime consumption, savings, and welfare in

[2]See De Nardi et al. (2016b) for an excellent survey.

[3]Some examples on precautionary savings motives include works by Hubbard et al. (1995), Palumbo (1999), De Nardi et al. (2010) and Kopecky and Koreshkova (2014a), and those on bequest motives include Hurd (1989), De Nardi (2004), De Nardi et al. (2010), Lockwood (2018) and Ameriks et al. (2011).

[4]For example, see Browning et al. (1994), Lundberg et al. (1997), Blundell et al. (2007), and Lise and Seitz (2011).

old age.

In addition, our paper contributes to the growing literature on home equity in retirement. Venti and Wise (2004) found that retirees typically do not liquidate home equity to support general nonhousing consumption unless they experience the death of a spouse or enter into a nursing home. Using an estimated life-cycle savings model, Nakajima and Telyukova (2020) found that homeowners dissave more slowly than renters because they have a preference for staying in their own home as long as possible and cannot easily borrow against it. A recent work by McGee (2019) used UK data to estimate a retirement savings model that incorporates house price shocks. Greenhalgh-Stanley (2012) found that Medicaid estate recovery programs reduce homeownership among single retirees, and Achou (2023) studied, in the context of single retirees only, the value of Medicaid's homestead exemption. We contribute to this literature by studying how asset composition interacts with intrahousehold dynamics and policy design to shape gendered welfare effects within married couples in retirement.

Finally, our work is related to the literature on old-age caregiving. Papers by Barczyk and Kredler (2018), Ko (2022) and Mommaerts (2025) use an intergenerational life-cycle savings model to study long-term care arrangements between elderly parents and adult children. A recent work by Barczyk et al. (2023) studies how housing assets can be used by parents as a commitment device to leave larger bequests and to elicit caregiving behaviors from their children. While our model accounts for care by adult children in a reduced-form manner, our contribution lies in endogenizing spousal caregiving, which is the primary source of family insurance for married retirees and therefore an important determinant of their precautionary savings motives.

2 Data and descriptive analysis

2.1 Data

The main dataset for this paper comes from the Health and Retirement Study (HRS) which has surveyed a representative sample of Americans over the age of 50 every two years since 1992. We use biennial interview waves from 1998 to 2014. We only consider individuals who were responsive and aged 65 and older in 1998; retired by 2000; and did not miss any interviews while alive.[5] The main empirical sample consists of 6,475 respondents and 35,865

[5]Respondents who move into a nursing home remain in the sample. We exclude individuals that leave the sample for reasons other than death to avoid conflating economic behavior with selective survey attrition.

respondent-wave observations. Appendix A provides details about the data, and Table G.1 provides summary statistics of the sample.

2.2 Construction of permanent income (PI)

We closely follow De Nardi et al. (2025) in specifying household PI. We assume that log household income evolves according to

$$\ln y_{it} = \kappa(t, f_{it}) + h(I_i) + \omega_{it}, \tag{1}$$

where i indexes household, $\kappa(t, f_{it})$ is a flexible function of household head's age t and family structure f_{it} (i.e., couple, single man, or single woman), $h(I_i)$ is specified as a flexible polynomial in I_i, and ω_{it} captures measurement error. From the estimated fixed effects, we construct a PI measure $\hat{I}_i$, which represents the rank order of each household's estimated fixed effect, and the corresponding estimate $\hat{h}(\hat{I}_i)$. As in De Nardi et al. (2025), this household's PI measure is invariant to changes in the family structure.

For our subsequent analysis, we also require a measure of the wife's share of household income among couples. To do so, we regress the household-level PI measure, $\hat{h}(\hat{I}_i)$, on the average incomes of the husband and the wife over the sample period. Using the estimated coefficients from this regression, we define the wife's income share for a household as the fraction of predicted household $\hat{h}(\hat{I}_i)$ attributable to the wife's average income. The mean value of the wife's income share across households is 0.34.

When we consider a finite number of PI values, we partition households into three groups based on terciles of $\hat{h}(\hat{I}_i)$. We refer to these groups as low PI, mid PI, and high PI, corresponding to the first through third terciles. We likewise form three groups for the wife's income share based on terciles of its sample distribution. Additional details are provided in Online Appendix B.

2.3 Evidence on collective decision making in retirement

We first examine whether married retirees' consumption and savings behaviors are consistent with a unitary model, a widely used framework in the literature. If unitary models provide a good approximation of couples' decision-making during retirement, then distribution factors, i.e., variables that influence the intrahousehold bargaining process but do not directly affect preferences or the budget constraint, should have no effect on household behavior.

Table 1: Wife's income share and household consumption-savings

	(1) Consumption rate	(2) Log of consumption	(3) Savings rate	(4) Change in assets
Wife's income share	-0.34**	-0.29***	1.22**	78361.31***
	(0.16)	(0.11)	(0.59)	(28889.56)
Observations	1703.00	1703.00	11012.00	11012.00
Adjusted R^2	0.40	0.42	0.12	0.18
Mean of dependent variable	1.39	10.87	-0.27	-13749.23

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the household level (in parentheses). Sample consists of married respondents from HRS 1998-2014. Consumption questions were asked only to a randomly selected subsample. See Table G.3 in the Online Appendix for coefficients on other covariates.

We use the wife's income share, defined in Section 2.2, as the distribution factor. This choice follows a large body of work that employs relative earnings as a distribution factor for working-age couples (e.g., Browning et al. (1994), Lise and Yamada (2019), and Lise and Seitz (2011)). While numerous studies have tested and rejected the income-pooling implication of unitary models for the working population, evidence for retired couples remains limited.[6]

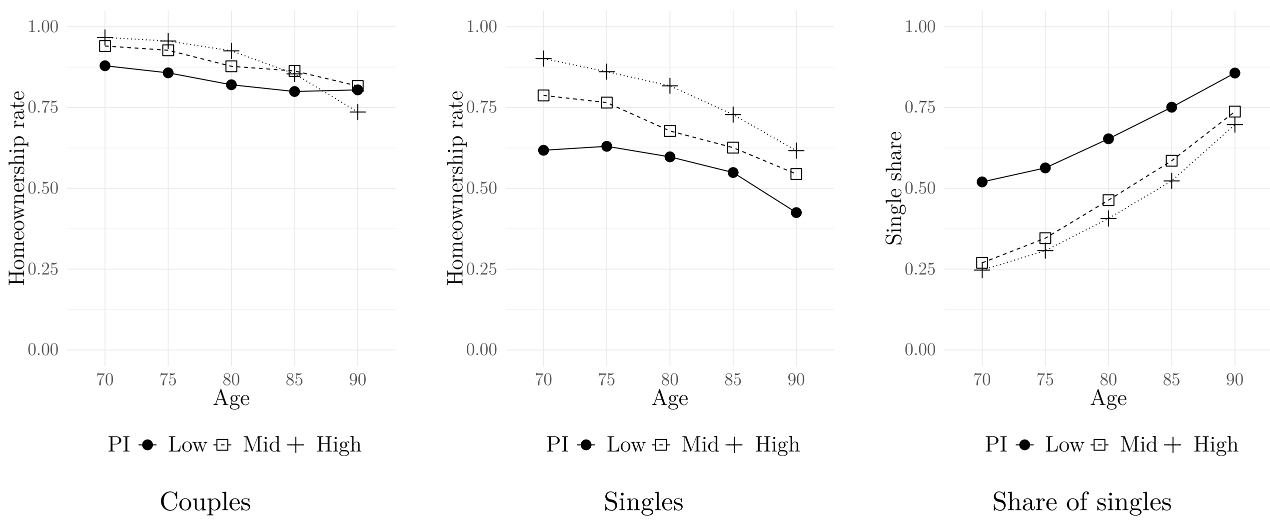
Table 1 reports the association between the wife's income share and household consumption-savings, conditional on household-level observables, individual-level characteristics of each spouse, and birth cohort fixed effects.[7] Consumption is measured using the HRS Consumption and Activities Mail Survey (CAMS) subsample, which consists of HRS respondents randomly selected and surveyed about their consumption behaviors. It is defined as the sum of all consumption in the household, including durable consumption, housing consumption, transportation consumption and nondurable spending. Column (1) uses the consumption rate (consumption to income ratio), and Column (2) uses log consumption. The results show that household consumption decreases with the wife's income share: 10–percentage point increase in the wife's income share is associated with a 3.4 percentage point decline in the consumption rate and a 2.9% decrease in consumption. The results are consistent with stronger precautionary saving motives among women, reflecting their longer life expectancy. Columns (3) and (4) use savings, defined as the change in total assets (liquid and housing) between waves. Column (3) uses the savings rate (savings to income ratio), and Column (4) measures savings in dollars. In line with Columns (1) and (2), they show that on average, the wife's income share is positively correlated with household savings.[8]

[6] See Chiappori and Mazzocco (2017) for an excellent review of the literature.

[7] See Table G.3 in the Online Appendix for the full list of controls.

[8] The lower adjusted R^2 for savings-based regressions likely reflects that savings are constructed as first differences of measured wealth, which amplifies potential measurement error in the data.

Figure 1: Homeownership rates by marital status



Notes: The figure reports homeownership rates by current marital status, PI and age.

In Section 4, we revisit this relationship using auxiliary regressions for the indirect inference estimation, which include a smaller set of controls limited to variables incorporated into our structural model and disaggregate the impact of the wife's income share by PI group. The results again provide evidence that household behavior responds to individual income shares. Together, these results provide descriptive evidence consistent with a collective model of household behavior in retirement, in which the wife's bargaining power rises with her relative income share.

2.4 Marital status and savings in retirement

We next document a series of descriptive patterns that guide the modeling choices for our collective framework, which explicitly incorporates marital transitions and housing. The recent study by De Nardi et al. (2025) provides detailed evidence on how total savings, combining liquid and housing assets, differ between couples and singles. We build on this literature by separately examining housing assets by marital status and by analyzing singles' consumption-savings behavior by gender.

Figure 1 presents homeownership rates by current marital status, PI terciles, and age. Figure G.1 in the Online Appendix further conditions on birth cohort. Singles are individuals who are not currently married, including widows and widowers, the never married, and the divorced. Because individuals transition from married to single through spousal death and exit the sample upon death, the data are not balanced. The third column reports the share

Table 2: Long-term care arrangements by marital status

	Married		Single
	Spouse healthy	Spouse disabled	
	(1)	(2)	(3)
Caregiving by spouse	0.75	0.39	—
:*by wife*	0.76	0.41	—
:*by husband*	0.74	0.36	—
Caregiving by children	0.25	0.49	0.63
Nursing home care	0.26	0.24	0.38
Observations	1366	397	4553

Notes: The sample consists of retirees that have long-term care needs. We classify individuals as having long-term care needs if they report at least two limitations in activities of daily living (ADLs) or exhibit cognitive impairment.

of singles among survivors at each age. Among married households, homeownership declines only modestly with age, with about 75% still owning homes at age 90. In contrast, singles exhibit rapid housing dissaving, and by age 90 only about half remain homeowners. Figure G.2 in the Online Appendix shows similar patterns using the median housing asset share. This stark divergence indicates that singles' faster dissaving is concentrated in housing: as Figure G.3 shows, non-housing asset trajectories are similar for couples and singles.

Exploiting the panel nature of the data, we conduct an event study analysis to examine changes in the homeownership rate around spousal death. Details are given in the Online Appendix C. The event study shows a substantial decline in homeownership following spousal death: relative to four years before the event, homeownership is about 12.5 percentage points lower two years after and 16 percentage points lower four years after.[9]

The descriptive patterns presented so far suggest that couples are more likely to hold housing assets, which leads to slower dissaving of retirement wealth. One possible reason is differential long-term care arrangements by marital status. Table 2 shows that for married disabled individuals with a healthy spouse, 75% receive spousal care. We find almost equal caregiving rates by wives (76%) and husbands (74%), with limited reliance on nursing homes (26%) or children (25%, mostly complementing spousal care). When both spouses are disabled, accounting for only 22% of married cases, care provided by children becomes most common (49%). For singles, care from children dominates (63%), and nursing home use is higher (38%) relative to couples. Lack of spousal care and increased risk of moving into a nursing home might contribute to singles' lower value for homeownership.

[9] These findings are consistent with earlier descriptive evidence on housing responses to spousal death. Using AHEAD data, Venti and Wise (2004) report that about 11 percent of initially homeowner couples exit homeownership following spousal death.

Table 3: Gender and consumption-savings among singles

	(1) Consumption rate		(2) Consumption rate		(3) Savings rate		(4) Savings rate	
Female	-0.14***	(0.04)	-0.26**	(0.11)	0.17	(0.11)	0.28	(0.18)
PI (mid) × Female			0.20*	(0.12)			0.20	(0.24)
PI (high) × Female			0.18	(0.12)			-0.68**	(0.33)
Observations	3537.00		3537.00		13711.00		13711.00	
Adjusted R^2	0.26		0.26		0.14		0.14	
Mean dep. var	1.08		1.08		-0.41		-0.41	

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the household level (in parentheses). Sample consists of single respondents from HRS 1998-2014. Consumption questions were asked only to a randomly selected subsample. See Table G.4 in the Online Appendix for coefficients on other covariates.

Another potential mechanism is Medicaid's differential treatment of housing assets in its means test by marital status. In the U.S., formal long-term care costs are costly, with median nursing home costs exceeding $90,000 annually in 2017. Medicaid is a means-tested public insurance program that finances long-term care for individuals with limited resources and ongoing care needs.[10] As the primary payer for long-term care in the U.S., Medicaid excludes the home from eligibility tests as long as it remains a "home." However, when an unmarried recipient moves into a nursing home without intent to return, Medicaid can place liens (TEFRA liens) on the property and recover long-term care costs from the estate upon death as a part of their Medicaid estate recovery program (Department of Health and Human Services, 2005). Greenhalgh-Stanley (2012) uses state-level policy variations and finds that such liens reduce singles' homeownership relative to couples. Using the HRS 1998-2014, which covers a longer and more recent period than Greenhalgh-Stanley (2012) and includes eleven states adopting the policy, we find that these liens reduce singles' homeownership by 8.9 percentage points relative to couples, confirming the negative effect documented in the earlier study.[11]

We conclude this descriptive section by documenting how consumption-savings behaviors differ by gender among single retirees. The sample for Table 3 consists of single retirees, and as in Table 1, we include a wide range of observables and birth cohort fixed effects.[12]

[10] Medicare, the primary public health insurance program for older adults in the U.S, covers only short-term stays in skilled nursing facilities following an acute hospitalization or illness. It does not cover long-term custodial nursing home care.

[11] Greenhalgh-Stanley (2012) exploits policy variation over 1993–2004 and uses the AHEAD cohort, which consists of individuals aged seventy and older in 1993 and follows them until 2004 or death. According to Greenhalgh-Stanley (2012), the homeownership rate for singles in TEFRA lien states decreases by 3 percentage points compared to couples.

[12] Table G.4 of the Online Appendix reports the full list.

Column (1) uses the consumption rate (consumption to income ratio) and indicates that single women consume less than single men; the result is robust to using log consumption as the dependent variable. This pattern is most pronounced in the lowest PI group and diminishes at higher PI levels, as shown in Column (2). Turning to savings behavior, Column (3) uses the savings rate (savings to income ratio) and suggests that women tend to save more on average, although the estimate is noisy. Allowing the relationship to vary with PI in Column (4) reveals substantial heterogeneity: in the low- and mid-PI groups, the estimates imply higher savings among women, although the effects are imprecisely estimated, while single women in the high-PI group save significantly less than their male counterparts.[13] Figure 2 in Section 4 includes the evolution of wealth among single retirees conditional on gender, age, PI, and birth cohort. It also indicates that among high-PI individuals, single females dissave wealth faster than their male counterparts.

3 Model

The model presented in this section describes retirees' consumption-savings and long-term care arrangement decisions in the face of health and mortality shocks. Time, $t = 1, 2, ..., T$, is discrete, and each period spans two years, consistent with the HRS's biennial interview schedule. In the first period, the household head is age 65, and the final period corresponds to age 99, after which individuals are assumed to die with certainty. In the initial period, households may be married couples, single females, or single males. For married couples, the husband is the household head, and we use a collective framework to describe couples' decision making process.[14] The model allows for gender heterogeneity not only in health and mortality risk, but also in preferences for caregiving, consumption-housing tradeoff, and bequest motives toward children. The model incorporates welfare programs including Medicaid as a lower bound on consumption. Individuals' discount factor is denoted by β.

[13] We confirm that the results are robust when savings in dollars is used as the dependent variable, although these estimates are less precisely measured.

[14] An alternative non-unitary framework is the limited commitment approach, in which large shocks to either spouse's outside option may cause participation constraints to bind, trigger changes in Pareto weights, and, if no feasible reallocation satisfies both constraints, lead to divorce. However, income shocks in retirement are small given the dominance of stable sources such as Social Security and pensions, and there were no major changes to divorce laws or Social Security rules during our sample period. As pointed out in Ligon et al. (2002) and Lise and Yamada (2019), limited commitment models make different predictions from collective models only when large income shocks occur. Consistent with small income shocks in retirement, divorce is exceedingly rare in the HRS: only about 0.5% of married retirees divorce over the 1998–2014 period.

3.1 Timing

At the start of each period, mortality shocks are realized. In the event of survival, health risks are first realized. If there is a sick member in the household, long-term care arrangements are determined which could be either spousal care (available only for married households), nursing home care, or informal care provided by adult children. Homeowners then decide whether to sell their home, and renters choose housing services. The government makes transfers to guarantee a minimum consumption floor. Finally, the household chooses consumption.

3.2 Preferences

Gender is denoted by $g \in \{W, M\}$, where W refers to women and M to men. For married households, we specify each member's preferences separately rather than using an aggregate household utility function. This methodological individualism is essential for analyzing intrahousehold inequality and aligns with neoclassical microeconomic analysis, which is based on individual preferences rather than those of a group (Chiappori, 1992). Flow utility in period t for a married individual of gender g is given by

$$U_t^g = u^g(c_t^g, h_t^g) - \psi^g P_t^g. \tag{2}$$

The function u^g represents gender-specific utility from non-housing and housing consumption. As is common in the literature (e.g., Lise and Seitz (2011), Blundell et al. (2005)), we allow for gender-specific preferences.[15] c_t^g denotes private non-housing consumption, and h_t^g represents housing services. Although housing is typically modeled as a public good (i.e., $h_t^g = h_t^{g'}$), we allow them to differ when one spouse resides in a nursing home, as such residency significantly constrains access to housing services. The indicator variable P_t^g equals 1 if individual g provides spousal care and 0 otherwise. Providing care yields a disutility of ψ^g. While ψ^g is expected to be positive, reflecting the psychic costs of caring for a disabled spouse, we allow it to take either sign.

For singles of gender g, the flow utility does not include the caregiving term, and it is given as $u^g(c_t, h_t)$. Following De Nardi et al. (2025), we assume that an individual's preferences over a given bundle of consumption do not depend on marital status.[16] This assumption also facilitates identification, as discussed in Section 4.2.1.

[15] When household members share identical preferences, collective models collapse to a unitary model. See Vermeulen (2002) for discussions about when unitary models emerge as a special case of a collective model.

[16] Voena (2015) also assumes consumption utility remains the same after getting divorced.

We adopt the functional form of

$$u^g(c_t, h_t) = \frac{{c_t}^{1-\gamma} - 1}{1-\gamma} + \sigma^g \frac{{h_t}^{1-\gamma} - 1}{1-\gamma}, \tag{3}$$

where c_t and h_t represent consumption of non-housing and housing goods, respectively. This functional form follows Campbell and Cocco (2007), who impose a common curvature on non-housing and housing consumption and introduce a relative weight on housing. Because identifying both the housing curvature and the housing preference weight is difficult with our data, we likewise use the same curvature for housing; further discussion is provided in Section 4.1. We introduce preference heterogeneity across genders through the housing preference weight σ^g, rather than through risk aversion. This choice is also motivated by identification considerations and is explained in detail in Section 4.1.

When there is no surviving spouse, the bequests go to children (or other heirs). As we do not model these agents' preferences, we adopt a warm-glow approach and specify bequest preferences as

$$v^{S,g}(b_t) = \delta^g \frac{(a^g + b_t)^{1-\gamma} - 1}{1-\gamma}, \tag{4}$$

where b_t denotes the total remaining household assets, including both liquid and housing assets. This is a commonly used functional form to model bequest motives when preferences of the inheritors are not specified (e.g., De Nardi (2004), De Nardi et al. (2010), and Lockwood (2018)). The parameter δ^g represents the intensity of the bequest motive, and a^g its curvature.

When a deceased individual is survived by a spouse, data from the HRS exit interviews show that the vast majority of bequests are left to the surviving spouse, with an average of 85.6% of the estate. We therefore assume that the surviving spouse retains the remaining assets. As the surviving spouse's preferences are specified, we model bequest motives of a married individual as pure altruism,

$$v^{C,g}(x_t) = \alpha V_t^{S,g'}(x_t), \tag{5}$$

where x_t is the household state vector, described in Section 3.10, and $V_t^{S,g'}(x_t)$ is the surviving spouse's continuation value as a single retiree, which depends on remaining household assets, among other elements. The altruism parameter α reflects the extent to which a married individual values the well-being of their surviving spouse at the time of their own death. While it may vary by gender, we estimate an average level for identification considerations, which we detail in Section 4.2.1. The use of pure altruism in modeling bequest motives is commonly used when the inheritor's preferences are specified (e.g., Kaplan (2012) and Ko (2022)).

An alternative way to specify bequest preferences upon the death of a married individual is to allow for bequests not only to the surviving spouse but also to children. However, as we describe in the Online Appendix D.1, identification becomes challenging in our collective framework. As such, we abstract from "side bequests" and focus instead on bequests to the surviving spouse, which dominate empirically and offer an opportunity to study within-household altruism at death, an aspect possible only in a collective framework.

3.3 Consumption

We assume non-housing consumption depends on nursing home (NH) residency:

$$c_t = \begin{cases} \hat{c}_t & \text{if not in NH,} \\ c_{nh} & \text{if in NH.} \end{cases} \tag{6}$$

When the individual is not in a nursing home, consumption is chosen and denoted by $\hat{c}_t$. In contrast, when the individual resides in a nursing home, consumption is fixed at a basic level c_{nh}. This reflects the institutional reality that nursing home services are delivered as a bundled package covering not only medical and housing services, but also meals, basic amenities, and routine assistance with daily activities, which substitute for the non-housing goods and services individuals would otherwise choose themselves. As a result, individuals have limited opportunities to enjoy additional consumption beyond what the facility provides. Papers such as Kopecky and Koreshkova (2014b) similarly model residents in publicly financed nursing homes as receiving a fixed level of consumption.[17]

The household consumption expenditure is given as

$$\chi_t = \begin{cases} [(\hat{c}_t^W)^\rho + (\hat{c}_t^M)^\rho]^{\frac{1}{\rho}} & \text{for couples,} \\ \hat{c}_t & \text{for singles.} \end{cases} \tag{7}$$

[17] In theory, the consumption value from nursing home care might depend on whether services are publicly or privately financed, as emphasized by Ameriks et al. (2011) and Ameriks et al. (2020). However, nursing home expenses are not separately observed in the HRS, which precludes modeling quality differences directly. Using the same HRS data, Lockwood (2018) finds that aversion to publicly financed care is modest: the consumption value (including housing) provided by publicly financed care facilities is estimated at $19,100 per year, which is very close to the $20,000 benchmark used for privately financed facilities.

$\rho \geq 1$ means there are economies of scale for couples' consumption. When none of the spouses is in a nursing home, then each spouse gets their own private consumption, denoted by $\hat{c}_t^g$ for $g \in \{W, M\}$. If only one spouse is in a nursing home, then that spouse's consumption expenditure is zero as they cannot consume beyond basic c_{nh}, and the other spouse gets the entire household consumption expenditure. If both spouses are in a nursing home, then the household will optimally choose $\chi_t = 0$.

3.4 Housing

Each family enters period t with initial housing assets denoted by $\tilde{h}_{t-1} \geq 0$. Strictly positive values of $\tilde{h}_{t-1}$ imply the household is a homeowner, and $\tilde{h}_{t-1} = 0$ means the household is a renter. Switching from a renter to a homeowner is rare in retirement, so we assume renting is an absorbing state.[18] Homeowners decide whether they sell home ($D_t = 1$) or keep it ($D_t = 0$). We do not allow homeowners to switch houses unless they become a renter, i.e., there is no downsizing. This assumption is grounded in the empirical study by Venti and Wise (2004), which finds that, on average, there is no reduction in housing equity among retirees who continue to own homes. The no-downsizing assumption has become standard in structural models that study housing decisions in retirement in the U.S.[19]

After homeowners' liquidation decision, housing assets are updated as the following:[20]

$$\tilde{h}_t = \begin{cases} \tilde{h}_{t-1} & \text{if } \tilde{h}_{t-1} = 0 \text{ or } (\tilde{h}_{t-1} > 0 \text{ and } D_t = 0), \\ 0 & \text{if } \tilde{h}_{t-1} > 0 \text{ and } D_t = 1. \end{cases} \tag{8}$$

Households with $\tilde{h}_t = 0$ decide on renting services, denoted by R_t.

Individual consumption of housing services is

$$h_t = \begin{cases} \omega_{f_t} \tilde{h}_t & \text{if not in NH and homeowner}, \\ R_t & \text{if not in NH and renter}, \\ h_{nh} & \text{if in NH}. \end{cases} \tag{9}$$

[18] In the HRS data, less than 10% of retirees purchase a new home.

[19] Nakajima and Telyukova (2020) assume that for homeowners, the housing decision is whether to move out of the house and rent or to stay in it. Achou (2023) similarly assumes that retirees do not switch homes unless they move to rental housing. Barczyk et al. (2023) adopt a related assumption, restricting home purchases beyond a certain age while allowing only sales.

[20] The model abstracts from housing price fluctuations and interprets saving behavior as arising from precautionary and bequest motives rather than from valuation changes in housing wealth. Explicitly modeling housing price risk, as in McGee (2019), is an important and complementary extension beyond the scope of this paper.

If a non-institutionalized individual is a homeowner, housing consumption equals the value of the home multiplied by the homeownership premium ω_{f_t}, which varies with marital status, f_t. Non-institutionalized renters derive utility from the rented housing service R_t. For individuals residing in nursing homes, housing consumption is fixed at h_{nh}, reflecting the standardized living arrangements provided by these facilities.[21]

Housing expenditure in each period is

$$e(\tilde{h}_t, R_t) = \begin{cases} \eta \tilde{h}_t & \text{if } \tilde{h}_t > 0, \\ (r+\eta) R_t & \text{if } \tilde{h}_t = 0, \end{cases} \tag{10}$$

where η is the depreciation rate, and r is the real interest rate. Liquidating housing assets worth of $\tilde{h}$ incurs transaction costs $\tau \tilde{h}$.

3.5 Household income

As described in Section 2.2, household income is a function of household head's age, family structure (f_t), and PI (I):

$$y_t = y(age_t, f_t, I). \tag{11}$$

3.6 Health and mortality risk

We consider three health statuses: $s_t \in \{\text{healthy, require long-term care, dead}\}$. Health transition probabilities follow a Markov chain. As in De Nardi et al. (2025), they depend on the individual's current health, age, gender, PI, and marital status,

$$Pr(s_{t+1} | s_t, age_t, g, I, f_t). \tag{12}$$

The health transition process is exogenous and does not depend on the receipt of care. This is based on previous studies which find that the evolution of elderly health is largely unaffected by the receipt of care (Finkelstein and McKnight, 2008; Card et al., 2008).

[21] As previously mentioned, this approach is consistent with Kopecky and Koreshkova (2014b), who model residents in publicly financed nursing homes as receiving a fixed level of consumption that includes housing. Lockwood (2018) finds only modest differences in the consumption value (including housing) between publicly and privately financed facility care.

3.7 Long-term care arrangements

When individuals become disabled, they require either informal or formal long-term care. To maintain model tractability, we assume that disabled individuals receive only one type of care at a time, as in Ko (2022) and Mommaerts (2025). For single individuals, the major source of long-term care is their adult children, as shown in Table 2.[22] Previous studies have shown that individuals generally prefer family-provided care over formal long-term care services (Barczyk and Kredler, 2018; Mommaerts, 2025; Ko, 2022; Barczyk et al., 2025). Based on these findings, we assume that disabled singles will rely on care provided by their children if it is "available" ($ic_{child} = 1$) and will resort to formal care only when it is not ($ic_{child} = 0$).[23] We abstract from the possibility that children help their parents pay for formal care services, as such financial transfers are negligible in the data.[24]

For married households, spousal care becomes the major source of informal care. If one spouse becomes disabled and the other spouse remains healthy, the household collectively decides whether the healthy spouse will provide spousal care, in which case no other type of care is needed. Empirically, 75% of such households rely on spousal care, as shown in Table 2, with almost equal care provision rates by gender. If the household decides that the healthy spouse will not provide care, then the disabled spouse will use formal care, rather than relying on care from children. This assumption is grounded in the finding that when the healthy spouse does not provide care to their disabled partner, the use of formal care services is more substantial than the help received from children.[25] Because the alternative to spousal care is formal care, spousal care helps avoid costly long-term care expenditures, thereby benefiting the household. The care provider experiences disutility, denoted ψ^g, which reflects forgone leisure and other non-monetary costs associated with caregiving.

When both spouses become disabled, which is relatively rare due to differences in the onset of long-term care needs by gender, care from children empirically becomes the major source

[22] Papers such as Barczyk and Kredler (2018), Ko (2022), and Mommaerts (2025) consider children as active agents in long-term care arrangement decisions. While endogenizing children's care decisions would be ideal, we adopt a more reduced-form approach, as our focus is on the intrahousehold dynamics of a couple.

[23] The HRS survey asks, "Suppose in the future, you needed help with basic personal care activities like eating or dressing. Will your daughter/son be willing and able to help you over a long period of time?" If the answer is positive for any of the respondent's children, we assume informal care from children is available ($ic_{child} = 1$). Otherwise, we assume it is not ($ic_{child} = 0$), and the individual uses nursing home care whenever they become sick. We have verified that the reported beliefs align reasonably with actual informal care receipt from children.

[24] Among disabled parents who use formal long-term care, the mean annual financial transfer received from their children is just $300 (Ko, 2022).

[25] Among couples in which a healthy spouse does not provide care to the disabled spouse, the extensive usage rate is 24% for nursing homes and 19% for care from children. On the intensive margin, children provide an average of only 4 hours of care per day, whereas nursing homes provide full-time care.

of assistance, as indicated in Table 2. As with singles, we assume that the disabled couple will rely on care provided by their children if it is available. Otherwise, they resort to formal care.[26]

We focus on nursing home care as the sole form of formal long-term care, based on the following grounds. First, nursing homes are the largest financial long-term care risk faced by the elderly: Kemper et al. (2005/2006) find that a person turning 65 in 2005 can expect to spend about $47,000 on formal care before death, most of which is on facility care at about $39,000. Second, the lion's share of Medicaid long-term care spending (about 70%) is attributed to nursing homes. Third, Medicaid's asset test rules regarding housing change dramatically by marital status for nursing home residency, but not for paid home care use.

3.8 Medicaid and government transfers

Government guarantees a minimum consumption floor through means-tested welfare programs such as Medicaid and Supplemental Security Income. To simplify notations, we define the household's cash-at-hand before government transfers as

$$\tilde{a}_t = a_t + y_t + \underbrace{I[D_t = 1](1-\tau)\tilde{h}_{t-1} - e(\tilde{h}_t, R_t)}_{\text{net proceeds from housing decisions}} - \underbrace{m_t,}_{\text{cost of NH}} \tag{13}$$

where $D_t = 1$ means the household sells home, and $e(\cdot,\cdot)$ is the housing expenditure. m_t represents the cost of nursing home care and is strictly positive only for nursing home users.[27]

The amount of government transfers that singles receive is

$$tr_t = \begin{cases} \max\{0, \bar{a}_{nh=0} - \tilde{a}_t\} & \text{if single and not in NH,} \\ \max\{0, \bar{a}_{nh=1} - (\tilde{a}_t + (1-\tau)\tilde{h}_{t-1})\} & \text{if single and in NH.} \end{cases} \tag{14}$$

For non-institutionalized singles, the government offers the homestead exemption by excluding housing assets from the eligibility test and makes transfers to ensure the consumption

[26] Table 2 shows that spousal care is observed even among couples in which both spouses are disabled. Yet the care hours are minimal, amounting to only about one-third of what a healthy spouse provides.

[27] We abstract from private long-term care insurance which pays for formal long-term care costs before Medicaid. In the data, the private insurance ownership rate is less than 10%. Furthermore, we focus on long-term care expenditure risk because the HRS does not provide a credible measure of acute medical expenses net of Medicare and before Medicaid transfers, information that is essential for a model in which savings, and therefore Medicaid eligibility, are endogenously determined. In addition, descriptive evidence shows that nursing home costs overwhelmingly drive out-of-pocket medical spending risk in old age, making long-term care the relevant margin for our structural analysis.

floor of $\bar{a}_{nh=0}$. For singles in a nursing home, their post-sales housing assets are counted.[28] Eligible single homeowners must first liquidate their home to pay for nursing home care, and then the government makes transfers to ensure the consumption floor of $\bar{a}_{nh=1}$.[29]

The amount of government transfers that couples receive is

$$tr_t = \begin{cases} \max\{0, 1.5\bar{a}_{nh=0} - \tilde{a}_t\} & \text{if couple and none in NH,} \\ \max\{0, \bar{a}_{nh=0} + \bar{a}_{nh=1} - \tilde{a}_t\} & \text{if couple and one in NH,} \\ \max\{0, \bar{a}_{nh=1} - \tilde{a}_t\} & \text{if couple and both in NH.} \end{cases} \tag{15}$$

The consumption floor for non-institutionalized couples is assumed to be 1.5 times the consumption floor for non-institutionalized singles, $\bar{a}_{nh=0}$, as in De Nardi et al. (2025).[30] The key difference from equation (14) is that housing assets are disregarded by Medicaid when a married individual ends up in a nursing home. This is consistent with the Medicaid rules which provide the homestead exemption to married households in a nursing home, as explained in Section 2.

3.9 Asset accumulation law

Non-housing assets evolve according to

$$a_{t+1} = (1 + r)(\tilde{a}_t + tr_t - \chi_t), \tag{16}$$

where χ_t is the household consumption expenditure described in equation (7). We assume there is no borrowing. Furthermore, we abstract from reverse mortgages which enable households to borrow against their home equity. The choice is based on the empirical fact that almost no retirees use such loans: according to Nakajima and Telyukova (2017), only 1.9% of eligible homeowners were reported to use reverse mortgages in 2013.

[28] To keep the computation tractable, we do not model different states and assume the Medicaid estate recovery program is effective. As of 2014, estate recovery programs were implemented in all 50 states, plus the District of Columbia.

[29] As nursing home residents already receive basic food and housing from the facility, we have $\bar{a}_{nh=1} < \bar{a}_{nh=0}$.

[30] Later, we calibrate $\bar{a}_{nh=1}$ to be zero as in Lockwood (2018). Hence, the consumption floor for institutionalized couples would be zero regardless of the constant in front of $\bar{a}_{nh=1}$ in equation (15).

3.10 Recursive formulation

This section provides a recursive formulation for a couple's problem. We derive that of singles in the Online Appendix D.2. A married household's state vector at the beginning of period t, after health shocks are realized, is

$$x_t = (a_t, \tilde{h}_{t-1}, s_t^W, s_t^M; I, ic_{child}), \tag{17}$$

where a_t is the non-housing wealth, $\tilde{h}_{t-1}$ is the housing wealth, and s_t^g is the health status of each spouse, $g \in \{W, M\}$. Time-invariant state variables are PI (I) and the availability of informal care from children (ic_{child}).

The household's choice vector is

$$q_t = (\hat{c}_t^W, \hat{c}_t^M, D_t, R_t, P_t^W, P_t^M), \tag{18}$$

where $\hat{c}_t^g \geq 0$ is private non-housing consumption for member $g \in \{W, M\}$, $D_t \in \{0, 1\}$ represents the house selling choice relevant only for homeowners, $R_t \geq 0$ is the rent choice relevant only for non-homeowners, and $P_t^g \in \{0, 1\}$ indicates spousal care provision of member $g \in \{W, M\}$.

Denote the survival probability of member g in period t by p_t^g, which also depends on current health, PI, and marital status, but is suppressed for notational simplicity. The couple's value function is given as

$$\begin{aligned}
V_t^C(x_t) \quad &= \max_{q_t} \quad \lambda(I,z)U_t^W + (1-\lambda(I,z))U_t^M \\
&+\beta p_t^W p_t^M E(V_{t+1}^C(x_{t+1})|x_t, q_t) \\
&+\beta p_t^W (1-p_t^M)(\lambda(I,z) + \alpha(1-\lambda(I,z)))E(V_{t+1}^{S,W}(x_{t+1})|x_t, q_t) \\
&+\beta(1-p_t^W)p_t^M(\alpha\lambda(I,z) + (1-\lambda(I,z)))E(V_{t+1}^{S,M}(x_{t+1})|x_t, q_t) \\
&+\beta(1-p_t^W)(1-p_t^M)E(\lambda(I,z)v^{S,W}(b_{t+1}) + (1-\lambda(I,z))v^{S,M}(b_{t+1})|x_t, q_t)
\end{aligned} \tag{19}$$

subject to budget constraints. The Pareto weight $\lambda \in (0, 1)$ reflects the relative bargaining power of the wife. Collective models assume that these weights vary with the environment, capturing cross-sectional differences in intrahousehold bargaining power (Chiappori, 1988, 1992).[31] For the working population, these weights are modeled as functions of wages and

[31] Cross-sectional variation in Pareto weights is essential for a non-trivial collective model. If the weight is fixed, the model assumes that intrahousehold bargaining power does not depend on prices or the environment, making it indistinguishable from a unitary framework. See Vermeulen (2002) for a discussion of when the unitary model becomes a special case of the collective model.

household non-labor income. Since we focus on retired households, we exclude wages but include PI. In addition, we allow the Pareto weights to depend on the wife's income share, z. This variable serves as a distribution factor, which refers to variables that affect the Pareto weights but not preferences nor the budget set (Browning et al., 1994). As shown in papers like Chiappori et al. (2002) and Blundell et al. (2005), the presence of distribution factors is crucial for identification, as we elaborate in Section 4.2.1. The use of relative retirement income as a distribution factor is not only supported by our descriptive results in Section 2.3, but also aligns with a large body of work that employs relative earnings as a distribution factor in studies of the working-age population, including Browning et al. (1994), Lise and Yamada (2019), and Lise and Seitz (2011).

When both spouses are alive in the next period, then the continuation value is that of a couple, where the expectation operator is taken with respect to health statuses of the next period. When the husband dies and the wife survives, then the husband cares about the wife's continuation value. If the altruism parameter satisfies $\alpha = 1$, then the couple's continuation value is simply the surviving wife's value as a single, $V^{S,W}$. On the other extreme, if the altruism parameter is zero, then the couple places only the wife's Pareto weight, λ, on her value as a single retiree in case of the husband's death. Similarly, when the wife dies and the husband survives, the couple values the husband's value function as a widower, $V^{S,M}$, using a weight that combines the husband's Pareto weight, $1 - \lambda$, and the wife's weight, λ, multiplied by her altruism toward her surviving husband, α. If both spouses die, then remaining household assets go to their children or other heirs, where $b_{t+1} = a_{t+1} + (1 - \tau)\tilde{h}_t$, in which case each member derives warm-glow utility according to $v^{S,g}$.

The model is solved backwards. For $t = T$, the terminal value for the couple is given as

$$V_T^C(x_T) = \lambda(I, z)v^{S,W}(b_T) + (1 - \lambda(I, z))v^{S,M}(b_T). \tag{20}$$

For $t < T$, with singles' value functions in hands (details in Online Appendix D.2), we can solve the couple's problem and obtain V_t^C.[32]

For our analysis of intrahousehold inequality, it is useful to define each married individual's value function. For $g \in \{W, M\}$, given a sequence of choices, let $V_t^{C,g}$ denote the individual value from being married. Their terminal values are given as

$$V_T^{C,W}(x_T) = v^{S,W}(b_T), \tag{21}$$

$$V_T^{C,M}(x_T) = v^{S,M}(b_T). \tag{22}$$

[32]Note that when neither spouse is in a nursing home, the aggregate non-housing consumption χ_t is allocated according to $\frac{\partial u^W(\hat{c}_t^W, h_t)/\partial c}{\partial u^M(\hat{c}_t^M, h_t)/\partial c} = \frac{1-\lambda}{\lambda}$ where $\hat{c}_t^M = (\chi_t^\rho - (\hat{c}_t^W)^\rho)^{\frac{1}{\rho}}$.

For $t < T$, given a sequence of choices, $V_t^{C,W}$ and $V_t^{C,M}$ are given as

$$\begin{aligned}
V_t^{C,W}(x_t) &= U_t^W + \beta p_t^W p_t^M E(V_{t+1}^{C,W}(x_{t+1})|x_t, q_t) + \beta p_t^W (1-p_t^M) E(V_{t+1}^{S,W}(x_{t+1})|x_t, q_t) \\
&+ \beta(1-p_t^W) p_t^M \alpha E(V_{t+1}^{S,M}(x_{t+1})|x_t, q_t) + \beta(1-p_t^W)(1-p_t^M) E(v^{S,W}(b_{t+1})|x_t, q_t) \\
V_t^{C,M}(x_t) &= U_t^M + \beta p_t^W p_t^M E(V_{t+1}^{C,M}(x_{t+1})|x_t, q_t) + \beta p_t^W (1-p_t^M) \alpha E(V_{t+1}^{S,W}(x_{t+1})|x_t, q_t) \\
&+ \beta(1-p_t^W) p_t^M E(V_{t+1}^{S,M}(x_{t+1})|x_t, q_t) + \beta(1-p_t^W)(1-p_t^M) E(v^{S,M}(b_{t+1})|x_t, q_t) \quad (23)
\end{aligned}$$

Note that at optimal allocations, we have $V_t^C(x_t) = \lambda(I,z) V_t^{C,W}(x_t) + (1-\lambda(I,z)) V_t^{C,M}(x_t)$.

4 Estimation

To estimate our model, we employ a two-step estimation procedure. In the first step, we set or estimate parameters outside the model. In the second step, we use indirect inference to estimate the remaining structural parameters within the model.

4.1 First-stage estimation

Health and mortality risks The model assumes that health transition probabilities follow an exogenously given Markov process, where next period's health status is determined by current health, age, gender, PI, and marital status. We estimate these transition probabilities using maximum likelihood with a flexible logit specification. Table 4 summarizes simulated life expectancy and long-term care risks at age 65. Life expectancy rises with PI, while years with disability decline with PI. Conditional on gender and PI, initially married retirees live longer than their single counterparts, whereas differences in long-term care years are negligible. Finally, the table indicates stronger precautionary saving motives for women: across PI groups and initial marital statuses, women live longer and spend more years in need of long-term care.

Risk aversion We fix the coefficient of relative risk aversion, γ, and do not allow it to differ by gender. This choice follows from the identification structure. Risk aversion and bequest motives are difficult to identify from wealth data alone, as both increase savings.[33] We therefore fix one and estimate the other. Because the literature provides stronger consensus on risk aversion among retirees than on bequest motives, we fix risk aversion and estimate

[33] As emphasized in De Nardi et al. (2016b), distinguishing risk aversion from bequest motives based only on wealth data is inherently difficult. Papers such as Lockwood (2018) bring in additional data (e.g., long-term care insurance demand) precisely to separate the two forces.

Table 4: Simulated life expectancy and years with disability at age 65

Initial marital status	Gender	PI=Low	PI=Mid	PI=High
A. Life expectancy (years)				
Married	Man	12.7	16.9	17.8
Married	Woman	17.9	22.0	22.8
Single	Man	11.0	14.7	15.8
Single	Woman	15.3	19.3	20.1
B. Years with disability				
Married	Man	2.4	1.7	1.4
Married	Woman	3.8	2.5	2.2
Single	Man	2.5	1.8	1.3
Single	Woman	4.1	2.6	2.2

Notes: The table reports simulated long-term care and life expectancy from age 65 onward, conditional on marital status at age 65, gender, and PI.

bequest motives.[34] Given the lack of empirical evidence on gender-specific risk aversion in retirement, we adopt the commonly used value of $\gamma = 3$ for both men and women.

We apply this same curvature to housing consumption, as shown in equation (3). Absent this restriction, housing curvature and the housing preference weight would both need to be identified from savings data, which is difficult because they can generate similar movements in the housing asset share. We therefore follow Campbell and Cocco (2007) and impose the same curvature for housing and non-housing consumption.

Pareto weight inputs We consider three values of PI, low, mid, and high, based on the terciles of the estimated distribution of households' fixed effects, as described in Section 2.2. We parameterize the Pareto weight as

$$\lambda(I, z) = \frac{\exp(\beta_0 + \beta_I \ln(\bar{y}(I)) + \beta_z z)}{1 + \exp(\beta_0 + \beta_I \ln(\bar{y}(I)) + \beta_z z)}, \tag{24}$$

which is similar to the specification in Lise and Seitz (2011), who model the wife's share of household full income for working-age couples.[35] For each PI group, we set $\bar{y}(I)$ as the median income among married households as predicted by equation (11). We also consider three values of the wife's income share (z), calculated as the median within each tercile of its

[34] For example, simulation papers such as Hubbard et al. (1995), Engen et al. (1999), Scholz et al. (2006), and Brown and Finkelstein (2008) all use a value of 3 for the coefficient of relative risk aversion. Structural estimates are typically in a similar range; De Nardi et al. (2016a) estimate 2.83, De Nardi et al. (2010) 3.84, De Nardi et al. (2025) 3.7, and Lockwood (2018) 4.5. In contrast, estimates of bequest motive strength vary more widely, which we discuss in Section 4.2.3.

[35] In their static collective framework, they specify the sharing rule of the full income as $\frac{\exp(x'\psi)}{1+\exp(x'\psi)}$, where the vector x contains the wife's potential share of household earnings, the log of full household income, and the spousal age gap.

distribution. The resulting representative values are 0.17, 0.30, and 0.44. The parameters $(\beta_0, \beta_I, \beta_z)$ are estimated internally which we describe in the next section.

Economies of scale The OECD modified equivalence scale assigns a value of 1 to the household head and 0.5 to the spouse. Based on this, we set the parameter on economies of scale in consumption for couples at 1.5. We set the homeownership premium for married couples so that the same magnitude of economies of scale applies to consumption and housing services. Given this condition, we set married couples' homeownership premium at 2.725 and that of singles at 2.162 so that the average homeownership premium is 2.5 as in Nakajima and Telyukova (2020).[36]

Other values We set the per-capita consumption floor for nursing home residents to zero ($\bar{a}_{nh=1} = 0$) following Lockwood (2018). We set the consumption value and housing value of nursing home services to be 40% each of the consumption floor for non-nursing home residents, $\bar{a}_{nh=0}$, which we estimate in the second stage. We set the transaction cost of selling a house at 7% of the house value, following Gruber and Martin (2003). We set the depreciation rate for housing assets at 1.7% per year following Nakajima and Telyukova (2020). As in Brown and Finkelstein (2008), we use 3% time preference rate per year and 3% annual real interest rate. For formal care prices, we use the average rates in 2008, which were $230 per day for nursing home care (MetLife, 2008).

4.2 Second-stage estimation via indirect inference

We estimate the remaining model parameters using indirect inference, targeting auxiliary models selected based on the identification arguments discussed below. Although the parameters are jointly identified from information contained in all auxiliary models, certain data patterns are especially informative about particular parameters. Our identification discussion highlights these key patterns to clarify the primary channels through which the data discipline the model.

4.2.1 Identification

Identification of Pareto weights In collective models, Pareto weights are generally not identified from preferences unless certain conditions are met. We begin by explaining how

[36]The lower premium for singles also reflects higher opportunity costs of homeownership, including a greater likelihood of moving into their children's homes and higher per-capita maintenance burdens (de Ruijter et al., 2005).

Table 5: Auxiliary regression I (empirical)

	(1) Consumption rate	
Wife's income share	-0.81**	(0.34)
PI (mid) × Wife's income share	0.62	(0.39)
PI (high) × Wife's income share	0.72*	(0.38)
Observations	1915.00	
Adjusted R^2	0.42	
Mean of dependent variable	1.24	

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the household level (in parentheses). This table reports the auxiliary regression estimated on the empirical data. For the auxiliary specification to be replicated on simulated data, the regression includes only covariates incorporated in the structural model, including ages, liquid and housing assets, health, availability of informal long-term care from children, household income, PI, and birth cohort fixed effects, whose coefficients are omitted from the table.

the presence of a *distribution factor* identifies variation in the Pareto weights with respect to their arguments, as shown in Blundell et al. (2005). Because we specify the Pareto weight as $\lambda(I, z)$, where I denotes PI and z represents the wife's income share, this amounts to identifying the marginal effects of I and z on λ.

Identifying the marginal effect of z is straightforward as income shares satisfy the definition of a distribution factor, meaning they influence household choices solely through their impact on bargaining power. Therefore, variation in household decisions across different income shares identifies how the Pareto weight responds to z. Specifically, we use how the household consumption rate changes in z, as captured by the baseline coefficient in the auxiliary regression reported in Table 5, which we further elaborate below.

Identifying the marginal effect of I is more challenging, because I enters both the budget and the Pareto weight. Our strategy again exploits the presence of a distribution factor. Specifically, we exploit how the effect of the distribution factor varies with PI, together with the parametric structure of $\lambda(I, z)$. Suppose we observe how the effect of z on the Pareto weight varies with I.[37] Then, by integrating this object with respect to z, we can recover the marginal effect of I on λ. The challenge is that we do not observe the interaction in the Pareto weight directly. However, as household choices depend on λ, their response to z and the way this response varies with I reflects the interaction in the Pareto weight.

To capture this relationship, we target the coefficients from an auxiliary regression of the household consumption rate on the interaction term between I and z. The key difference from the regression in Section 2 is that here we focus on how the effect of the distribution

[37] This corresponds to the cross-partial derivative, $\partial^2\lambda/\partial I \partial z$.

factor varies with PI, as the interaction term identifies the cross-partial effect of I and z. Furthermore, the regression includes only controls incorporated into the structural model so that it can be replicated using simulated data.[38] The empirical results in Table 5 show that household consumption declines with the wife's income share, consistent with Section 2, but this negative effect *weakens* as PI increases. Mechanisms such as means-tested social insurance would predict the opposite pattern: at low PI, the consumption floor would bind and mute the influence of distributional factors, while at high PI, the household becomes less constrained and more responsive to changes in bargaining power. Observing the reverse pattern instead suggests a countervailing force. Under our parametric assumption of $\lambda(I, z) = \frac{e(I,z)}{1+e(I,z)}$ where $e(I, z) = \exp(\beta_0 + \beta_I \ln(\bar{y}(I)) + \beta_z z)$, this pattern is consistent with $\beta_I < 0$. The inequality would imply the wife's bargaining power declines with household income, as found in Lise and Seitz (2011). Accordingly, the regression coefficients on the interaction terms provide key information for identifying β_I.[39]

We also identify the level of the Pareto weights, which amounts to identifying β_0. Earlier works on collective models focus on identifying only the derivatives of bargaining power, not its level. Studies that require the level of Pareto weights for inequality analysis usually impose a normalization, for example by assuming equal weights for certain households (e.g., Lise and Seitz (2011)). Some works have shown identification of the location of the Pareto weights by using combined data on singles and couples and assuming certain preferences do not change with marital status (e.g., Vermeulen (2006), Browning et al. (2013)). Similar to these studies, we assume the same preference over general and housing consumption between married and single individuals, as shown in equation (3). In addition, we exploit variation in household decisions arising from the nursing home entry of one spouse. This event constrains the institutionalized spouse's consumption, as described in equations (6) and (9). As a result, the household's decisions in that period are more influenced by the preferences of the community spouse. Crucially, the change in behavior in response to the nursing home entry provides information about the pre-entry Pareto weights: if the community spouse's weight

[38] We use the consumption rate to remove the mechanical scaling effect of higher income. Empirical results using log consumption are qualitatively similar. We prefer consumption-based regressions because consumption is directly observed, whereas savings are inferred from first differences in wealth, which amplifies potential measurement error in the data. In this auxiliary specification, we find that savings-based regressions produce less precise estimates.

[39] Within our model, an alternative explanation for the pattern in Table 5 is that women have weaker bequest motives toward children when they die as single retirees. In that case, for high-PI households where bequests are more relevant, an increase in wives' bargaining power could raise consumption. However, gender-specific bequest motives when dying as a single retiree are primarily identified using singles' savings profiles conditional on gender as discussed below, which are free from intrahousehold dynamics and therefore discipline preference heterogeneity separately. Conditional on preference parameters that are disciplined by singles' savings profiles, the remaining variation in Table 5 primarily informs β_I, the parameter governing how bargaining power varies with PI.

Table 6: Auxiliary regression II (empirical)

	(1) Consumption rate	
Spouse enters NH x Entering spouse is male	-0.28	(0.17)
Observations	400.00	
Adjusted R^2	0.48	
Mean of dependent variable	1.32	

Notes: Standard errors are clustered at the household level (in parentheses). This table reports the auxiliary regression estimated on the empirical data. The regression also includes an indicator for spousal nursing home entry, ages, liquid and housing assets, health, availability of informal long-term care from children, household income, PI, wife's income share, and birth cohort fixed effects, whose coefficients are omitted from the table.

was high, the household would exhibit a relatively small behavioral change; in contrast, if the community spouse's weight was low, the adjustment would be more pronounced.

Based on this argument, we target the coefficients from an auxiliary regression that relates changes in household consumption to spousal nursing home entry. Specifically, we use a sample of couples who are observed both before and after nursing home entry of one spouse during marriage, and we exclude waves in which both spouses are in a nursing home. We regress the household consumption rate on an indicator for spousal nursing home entry and its interaction with an indicator for whether the entering spouse is the husband. The specification includes only variables incorporated into the structural model, ensuring it can be replicated using simulated data. The empirical coefficient on the interaction term is reported in Table 6.[40] The household consumption rate declines further by 0.28 when the husband enters the nursing home and the wife gains a greater influence.[41] We use this response to help discipline the level of the Pareto weights in the second-stage estimation.[42]

Identification of remaining parameters To help identify gender-specific preferences, we exploit variation by gender among single retirees. While couples' decisions reflect both preferences and relative bargaining power, singles' choices are not confounded by intrahousehold bargaining and therefore provide cleaner information about preference heterogeneity by gender. For instance, to discipline (σ^W, σ^M), the parameters governing consumption-housing tradeoff by gender, we use singles' homeownership rates conditional on gender and PI.

[40]Institutionalization restricts the entering spouse's consumption opportunities, and the main effect of spousal nursing home entry is largely driven by this mechanical component and is therefore less informative about intrahousehold bargaining power.

[41]The p-value is about 0.11.

[42]The result is consistent with wives having stronger precautionary savings motives. Because health and mortality risks by gender are set in the first stage of the estimation procedure, this auxiliary model is used to discipline the level of the Pareto weights in the second stage.

The parameters (δ^g, a^g) govern the bequest motives of gender $g \in \{W, M\}$ in the absence of a spouse. To help identify these parameters, we use the median total assets of singles conditional on gender, PI terciles, age groups, and birth cohort. Each age group spans four years, and birth cohorts are defined based on age in 1998.

While we allow for gender-specific bequest motives among singles, we assume a common spousal altruism parameter α for married individuals. This is due to identification considerations. Gender-specific bequest motives toward children can be identified from the wealth trajectories of single men versus single women.[43] However, such clean variation does not exist for couples, as we only observe household-level savings. We therefore estimate an average altruism toward the spouse. Couples' wealth profiles are particularly useful in identifying α, and we use the median total assets of couples conditional on PI terciles, age groups, and birth cohort.

To help identify the utility cost of providing spousal care, (ψ^W, ψ^M), we exploit spousal care rates conditional on gender. While these rates also crucially depend on relative bargaining power, the auxiliary models described above discipline the bargaining power channel, allowing spousal care patterns to be especially informative about the caregiving parameters.

To help identify the consumption floor, we use the Medicaid recipiency rates of couples, single men, and single women.

4.2.2 Estimation procedure

We implement indirect inference in a Bayesian way (Fernandez-Villaverde et al., 2016) using a multiple-block Metropolis–Hastings algorithm. Details are provided in the Online Appendix E. The algorithm splits the parameters into different blocks, as different subsets of parameters are primarily informed by different sets of auxiliary models, as described above. In simulations, we account for cohort heterogeneity by matching wealth moments across birth cohorts. To address survival bias, we incorporate mortality heterogeneity into the model and assign simulated households the same mortality histories observed in the data.

4.2.3 Estimation results

Table 7 reports the structural parameter estimates obtained via indirect inference. To the best of our knowledge, the literature provides no direct estimates of altruism toward surviving spouse. Most structural bequest models with couples either (i) treat the household as a single

[43] This identification strategy works given that we use the same risk aversion for both genders.

Table 7: Internally estimated parameters via indirect inference

Parameter	Description	Estimate	[90% CI]
α	Altruism toward surviving spouse	0.121	[0.083, 0.385]
δ^W	Bequest intensity, single women (10^6)	0.496	[0.470, 0.637]
a^W	Bequest curvature, single women (10^6)	12.481	[12.341, 12.493]
δ^M	Bequest intensity, single men (10^6)	0.253	[0.146, 0.262]
a^M	Bequest curvature, single men (10^6)	5.660	[5.130, 5.702]
σ^W	Weight on housing consumption, women	0.207	[0.081, 0.207]
σ^M	Weight on housing consumption, men	0.213	[0.149, 0.218]
ψ^W	Caregiving disutility, wives (10^{-8})	0.498	[0.498, 0.554]
ψ^M	Caregiving disutility, husbands (10^{-8})	0.179	[0.172, 0.190]
β_0	Pareto weight — level	3.557	[3.503, 3.557]
β_I	Pareto weight — PI	-0.606	[-0.625, -0.603]
β_z	Pareto weight — income share	9.061	[9.009, 9.333]
$\bar{a}_{nh=0}$	Annual consumption floor (10^3)	12.164	[12.116, 12.328]

Notes: We report posterior medians as point estimates, with 90% credible intervals in brackets.

decision maker with a joint bequest motive, or (ii) focus on transfers to children and abstract from a distinct altruistic component toward the surviving spouse. In contrast, our model delivers an estimated spousal altruism parameter (α) around 0.12. This is smaller than the altruism toward children typically found in the literature: for instance, Lee and Seshadri (2019) find 0.32, Gayle et al. (2022) obtain 0.80, and Caucutt and Lochner (2020) find 0.86.

The parameters governing bequest motives toward children (δ^g, a^g) differ meaningfully by gender. To interpret the magnitude, we follow the practice in the literature and compute the asset threshold below which individuals do not leave any bequests when death in the next period is certain. A lower threshold corresponds to stronger bequest motives. Under this metric, the threshold is $\underline{a}^W = \$78{,}842$ for women and $\underline{a}^M = \$44{,}767$ for men. Existing studies typically estimate bequest motives without allowing for gender heterogeneity, so the reported values reflect gender-average thresholds. These estimates span a wide range, from \$3,500 (De Nardi et al., 2016a) to \$43,000 (De Nardi et al., 2010), suggesting that our values for women are higher than what is typically found in the literature. The stronger bequest motive toward children that we find for men is consistent with descriptive patterns in Figure 2 and Table 3, where single men in the high–PI group decumulate their wealth at a slower pace than single women.

The estimated weights on housing consumption (σ^g) are similar across genders. Caregiving disutility (ψ^g) is positive for both genders, indicating that providing care to a disabled spouse is costly, despite being unrestricted in sign during estimation. While the estimates appear small, this reflects the high curvature of the CRRA utility function with coefficient three and the fact that the disutility enters additively. To interpret the magnitude, we compute for each gender, how much an individual with that gender's median non-housing consumption among married households would be willing to forgo to avoid providing care. For husbands, this value is $36,192 per year, while for wives it is smaller at $33,306, despite the higher estimate of ψ^W. This is because wives have lower median consumption and therefore higher marginal utility, so a given utility loss corresponds to a smaller dollar equivalent. These values fall within the range of monetary caregiving costs reported in the literature: Feinberg et al. (2011) estimate the annual cost of informal care at $10,678; Ernst and Hay (1994) calculate the net cost of informal care for an Alzheimer's patient to be $33,038 per year; and Skira (2015) estimates middle-aged women's cost of providing long-term care to their disabled mother at $82,363 per year.

Regarding the Pareto weight estimates, the negative value of β_I implies that wives' bargaining power decreases with PI. This is similar to Lise and Seitz (2011), who estimate a sharing rule within a collective model and find that women's consumption share decreases with household income. It is also consistent with Blundell et al. (2007), who show that as total resources increase, a larger share of the additional income accrues to the husband, thereby reducing the wife's relative share. The estimate of β_z is positive, suggesting that women's bargaining power increases with their income share. While the literature typically fixes the level of Pareto weights (β_0) by assuming equal weights for certain households, we estimate this value internally. We discuss the variations in Pareto weights across households in Section 5.1.

Our estimated annual consumption floor is approximately $12,164, close to the $12,000 found by Mommaerts (2025), but higher than the $4,100 estimated by De Nardi et al. (2025).

4.3 Model fit

Targeted objects Tables 8 and 9, together with Figure 2, summarize the fit of all auxiliary models used in the indirect inference estimation. The estimated model replicates most auxiliary regression coefficients that are informative for identifying the Pareto weights, although it overstates the effect of the wife's income share on lowering consumption for mid-PI households. The model matches Medicaid recipiency rates well for single women, but understates these rates for single men and couples. Figure 2 shows that the model provides a reasonable approximation to total wealth evolution conditional on birth cohort, current marital status, and PI.

Table 8: Fit of auxiliary regression coefficients

Dependent variable	Regressor	Data	Model
Auxiliary regression I			
Consumption rate	Wife's income share	-0.81	-0.77
Consumption rate	PI (mid) × Wife's income share	0.62	0.19
Consumption rate	PI (high) × Wife's income share	0.72	0.76
Auxiliary regression II			
Consumption rate	Spouse enters NH × Entering spouse is male	-0.28	-0.27

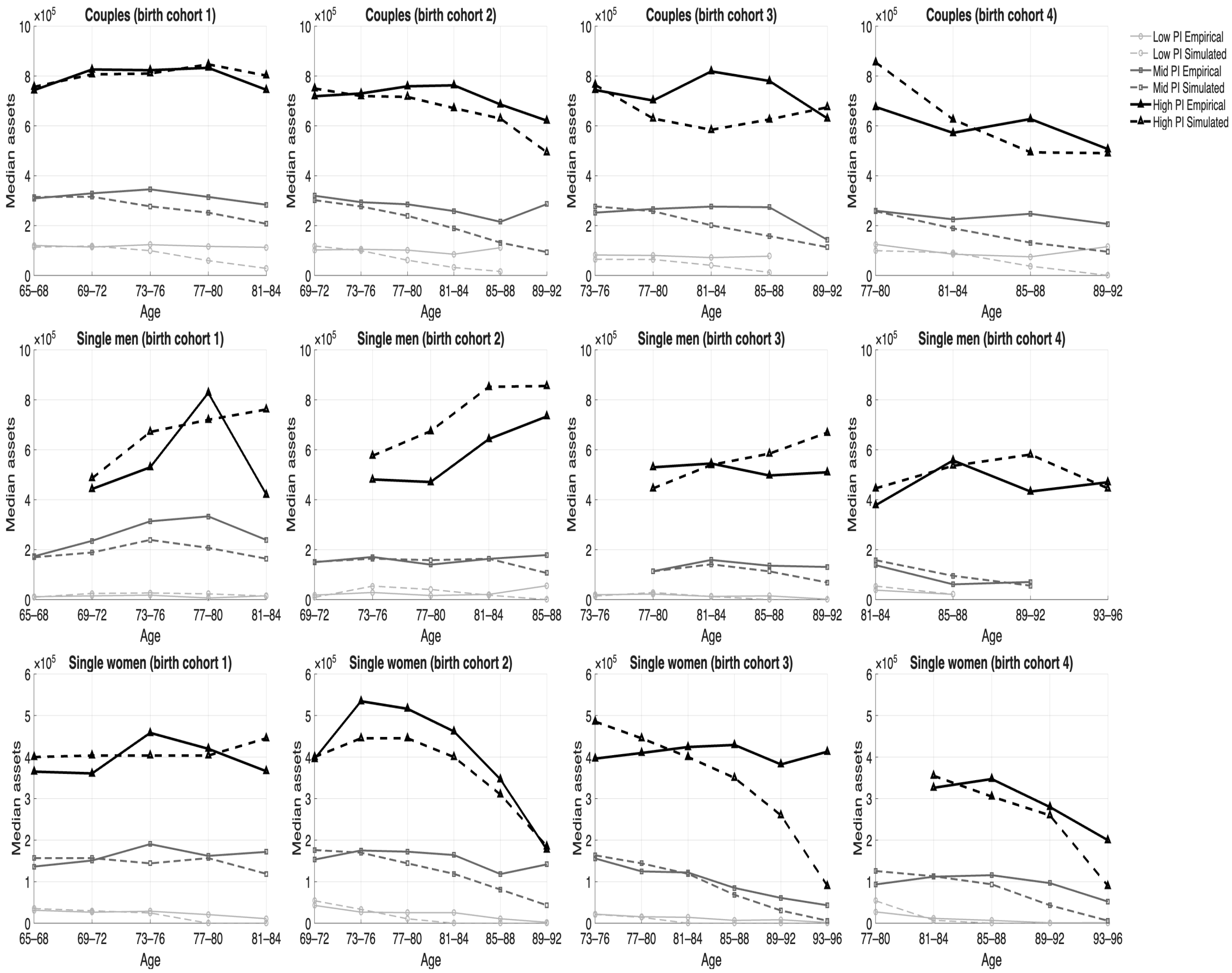
Table 9: Fit of auxiliary moments (rates)

Moment	Data	Model
Spousal caregiving rates		
By wife	0.76	0.66
By husband	0.74	0.87
Medicaid recipiency rate		
Couples	0.07	0.03
Single females	0.18	0.18
Single males	0.14	0.08
Homeownership rate of single females		
Low PI	0.47	0.20
Mid PI	0.64	0.48
High PI	0.76	0.70
Homeownership rate of single males		
Low PI	0.41	0.26
Mid PI	0.61	0.49
High PI	0.74	0.68

Untargeted objects For external validation, we also assess the model's fit for several untargeted statistics. For instance, regressing the consumption rate on the wife's income share (without conditioning on PI-group heterogeneity) on the simulated data yields a coefficient of –0.39, which is close to the empirical coefficient of –0.34 shown in Table 1 of Section 2. In Table 10, we also compare the model to empirical long-term care arrangements reported in Table 2 of Section 2. Some categories are intentionally omitted because they are not feasible within the model, which permits only one type of care at a time, as in Ko (2022) and Mommaerts (2025).[44] Even with this restriction, the estimated model performs reasonably well in matching care arrangements by marital status.

[44]These modeling assumptions are empirically motivated and discussed in Section 3.

Figure 2: Fit of auxiliary moments (wealth)



Notes: The figure reports empirical (solid lines) and simulated (dashed lines) wealth, defined as the sum of liquid and housing assets, conditional on birth cohort, current marital status, PI, and age. Darker colors indicate higher PI. Birth cohorts are defined based on household ages in 1998.

5 Main results

5.1 Intrahousehold inequality in bargaining power and lifetime consumption

This section uses the estimated model to quantify intrahousehold inequality in bargaining power and lifetime consumption among couples in retirement. We construct initial conditions from the 1998 and 2000 HRS waves using couples in which the head is aged 65–69, and then

Table 10: Fit of untargeted long-term care arrangements

	Married		Single
	Spouse healthy	Spouse disabled	
	(1)	(2)	(3)
Caregiving by spouse	0.75	—	—
	[0.75]		
Caregiving by children	—	0.55	0.63
		[0.49]	[0.63]
Nursing home care	0.25	0.45	0.37
	[0.26]	[0.24]	[0.38]

Notes: Entries without brackets report model-simulated rates. Entries in square brackets report empirical rates from Table 2 .

simulate their choices forward over the life cycle.[45] We construct a second set of initial conditions using couples with household heads aged 65–69 from the 2012 and 2014 waves, who were too young to be included in the estimation sample.[46] We run the simulations separately for each cohort and compare outcomes between couples entering retirement in 1998–2000 and those entering retirement in 2012–2014.

For each couple, we compute the wife's share of household consumption over the lifetime of the marriage using non-housing consumption, c_t^g, as in equation (2). We exclude housing consumption because it is mostly a public good.[47] We use the model-generated consumption paths for the husband and wife during periods in which both spouses are alive as a couple. We then compute the present-discounted value of each spouse's non-housing consumption over these periods. The wife's share is defined as the ratio of her present-discounted consumption to the household total.

The resulting measure captures the wife's consumption share over the life cycle of her marriage and is not observed in the data, which report consumption only at the household level. A collective model that maps household choices into individual consumption through relative Pareto weights is therefore essential for measuring intrahousehold consumption inequality over the life cycle.

Table 11 reports, for each cohort, the joint distribution of PI and wives' income shares, which together determine relative Pareto weights. As described in Section 2.2, we define the tercile cutoffs using the HRS estimation sample, which includes the older cohort shown in the left panel of Table 11. We then apply these same cutoffs to the younger cohort, shown

[45] While the model starts at age 65, we construct initial conditions using ages 65–69 to increase sample size and reduce noise.

[46] See Section 2.1 for the estimation sample construction.

[47] Including housing would mechanically dampen differences and obscure consumption inequality within the household.

Table 11: Composition of (PI, z) bins by cohort

Old cohort

	Low z	Middle z	High z	Total (PI)
Low PI	0.11	0.08	0.07	0.26
Middle PI	0.14	0.10	0.12	0.36
High PI	0.11	0.10	0.16	0.38
Total (z)	0.36	0.28	0.35	1.00

Young cohort

	Low z	Middle z	High z	Total (PI)
Low PI	0.08	0.06	0.10	0.25
Middle PI	0.10	0.05	0.12	0.27
High PI	0.15	0.09	0.25	0.48
Total (z)	0.33	0.20	0.47	1.00

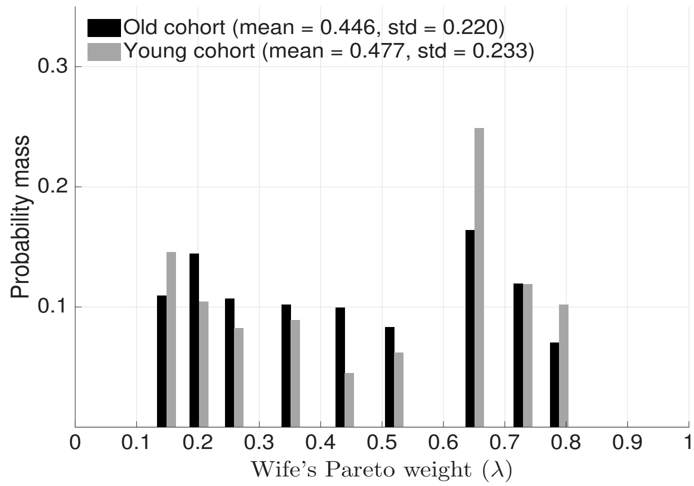
Notes: "Old cohort" consists of couples in which the husband was aged 65–69 in the 1998–2000 HRS waves. "Young cohort" consists of couples with husbands aged 65–69 in the 2012–2014 waves. Each panel reports shares in (PI, z) bins, where PI stands for permanent income and z denotes the wife's income share. The last column shows totals by PI and the last row shows totals by z.

in the right panel, to ensure consistent classification into three PI and three wife's income share groups. Across the two cohorts, the middle-PI share declines by 9 percentage points, with a corresponding increase in the high-PI share. A similar reallocation occurs for wives' income shares, with the middle-share group falling by about 8 percentage points and the high-share group rising by 12 percentage points.

Figure 3 plots the distribution of wives' Pareto weights for the two cohorts. In the earlier cohort, women's mean Pareto weight is 0.446, while in the later cohort it is 0.477. These values are higher than what is typically found for the working-age population. For instance, Voena (2015) finds approximately 0.30 for the wife's Pareto weight by leveraging employment responses to changes in U.S. divorce laws. The increase in the mean across the two cohorts reflects a greater share of couples in which wives have a higher income share and therefore larger bargaining power. In both cohorts, there is substantial dispersion, with the standard deviation in each cohort roughly one-half of the corresponding mean. The younger cohort exhibits a rightward shift in the distribution, along with slightly greater dispersion.

Figure 4 illustrates how differences in Pareto weights translate into consumption allocations over the life cycle. For each cohort, the distribution of wives' lifetime consumption shares closely mirrors the distribution of their Pareto weights, although dispersion in consumption shares is attenuated by consumption floors. For the older cohort, the mean is 0.482, implying that women who entered retirement in 1998–2000 received about 48.2% of household consumption over the lifetime of the marriage. For comparison, Lise and Seitz (2011) use a static model and estimate that in 2000, working-age women in childless U.K. couples had a consumption share of 0.40. Browning et al. (2013) find a mean wife's resource share of 0.63 using Canadian working-age couples with no other household members present. Cherchye et al. (2012) also find a mean wife's consumption share of 0.63 among Dutch retirees using

Figure 3: Distribution of wife's Pareto weight



Notes: The black bars plot the distribution of wives' Pareto weights for couples in which the husband was aged 65–69 in the 1998–2000 HRS waves ("Old cohort"). The gray bars show the corresponding distribution for couples with husbands aged 65–69 in the 2012–2014 waves ("Young cohort").

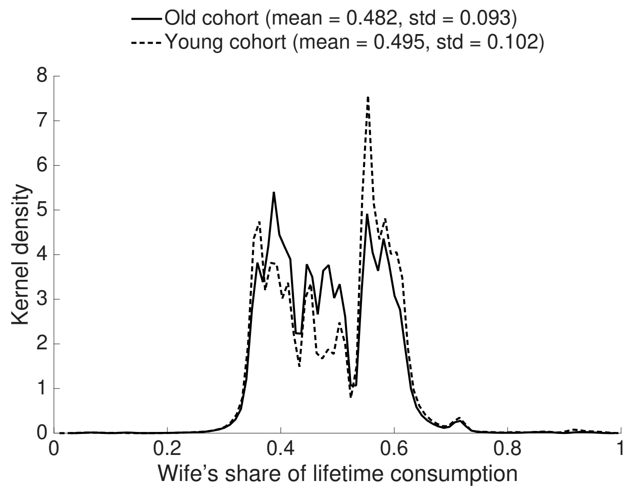
their static framework.[48] Figure 4 shows that for the younger cohort, the distribution shifts slightly to the right, with the mean rising to 0.495. As with the Pareto weights, the younger cohort exhibits slightly greater dispersion.

5.2 Intrahousehold heterogeneity in savings motives

We now examine heterogeneity in savings motives between spouses within a couple. While the existing literature has studied different savings motives, such as precautionary versus bequest motives, across household types and across the PI distribution, it has not been able to quantify intrahousehold heterogeneity in savings motives, which is infeasible under a unitary model. Our estimation results indicate meaningful gender differences in savings motives. Women exhibit stronger precautionary motives because they face longer longevity and higher disability risks, whereas bequest motives toward children upon dying as a single retiree, relevant primarily for high-PI retirees, are weaker for women. How these differences translate into couples' observed savings depends on the Pareto weights, which have an estimated mean of around 0.45 for women. To isolate each spouse's savings motives, we conduct counterfactuals in which either the wife or the husband receives the entire Pareto weight. This

[48] For studies on developing countries, women's relative resource shares range from 0.38 to 0.55 as reviewed in Jayachandran and Voena (2025).

Figure 4: Distribution of wife's share of lifetime consumption



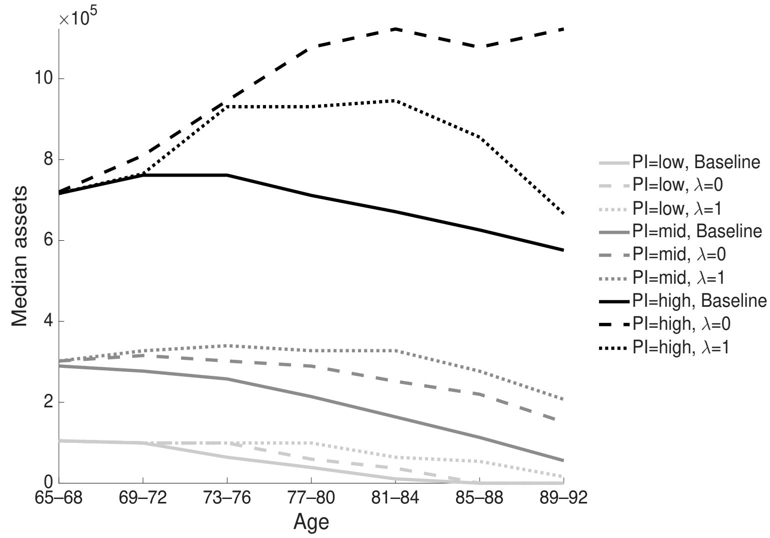
Notes: The solid line plots the distribution of the wife's share of consumption over the lifetime of the marriage for couples in which the husband was aged 65–69 in the 1998–2000 HRS waves ("Old cohort"). The dashed line shows the corresponding distribution for couples with husbands aged 65–69 in the 2012–2014 waves ("Young cohort").

counterfactual isolates how that spouse's preferences map into household savings decisions in the absence of intrahousehold compromise.

From the estimation sample, we construct initial conditions that include both initially married and initially single households, with a household head aged 65–69.[49] Figure 5 reports how median asset profiles, defined as the sum of liquid and housing assets, change under these counterfactuals for currently married couples conditional on PI and age. When either spouse receives all bargaining power, couples' savings increase across all PI groups relative to the baseline. Changes in spousal care provision and the resulting increase in long-term care expenditure risk are central to this pattern. In both counterfactuals with extreme Pareto weights, the spouse whose utility is fully weighted never provides spousal care to avoid the utility cost of caregiving. Although the unweighted spouse's care rate increases to one, the couple's overall reliance on formal long-term care services rises. As shown in Table 10, spousal care is an important source of insurance for both genders in the baseline. Therefore, when one spouse essentially shuts down care provision, the household's exposure to long-term care expenditures increases. This strengthens precautionary saving motives, especially for high-PI households whose probability of qualifying for means-tested Medicaid is low. Consequently, the increase in savings is the greatest in the high-PI group under both

[49]While the counterfactual pertains to changing relative Pareto weights and therefore does not affect individuals who are initially single, we include them to report how wealth accumulation changes for currently singles conditional on age and PI in Figure G.5 of the Online Appendix.

Figure 5: Counterfactual wealth trajectories for couples



Notes: The figure plots median wealth among currently married couples, defined as the sum of liquid and housing assets, by PI and age, with darker colors indicating higher PI. Solid lines show the baseline; dashed lines show the counterfactual in which the husband's Pareto weight is one ($\lambda = 0$); and dotted lines show the counterfactual in which the wife's Pareto weight is one ($\lambda = 1$).

counterfactuals.

Which counterfactual leads to a larger increase in savings varies across PI groups. For the low- and mid-PI groups, savings increase more when the wife holds all bargaining power (the dotted lines lie above the dashed lines). In these groups, bequests are less relevant, so precautionary motives dominate, which are stronger for women than men. As a result, when the wife's preferences are fully weighted, low- and mid-PI households save more than when the husband's preferences are fully weighted. In contrast, for high-PI households, savings rise more when the husband has all bargaining power (the dashed line lies above the dotted line). This pattern is driven by men's stronger bequest motives toward children, an effect that becomes more pronounced as the household ages.[50]

5.3 Distributional welfare effects of policies

This section considers the distributional welfare effects of reforms to old-age social insurance programs, focusing on Medicaid. Our measure of welfare is the compensating wealth

[50] In Figure G.5 of the Online Appendix, we report the corresponding results for individuals who are currently single. Because couples accumulate more assets under both counterfactuals, surviving partners enter singlehood with higher wealth, which mechanically leads to higher average assets among singles at older ages.

Table 12: Welfare effects of counterfactual Medicaid

	(1)	(2)	(3)	(4)
	Homestead exemption			Share
	Nobody	Everybody	Singles	
Initially couples	-658	38	-618	0.58
: *Wives*	-800	59	-798	
: *Husbands*	-569	-3	-502	
Initially single women	0	3777	3777	0.32
Initially single men	0	1785	1785	0.11
All (household-weighted)	-379	1410	1032	
All (individual-weighted)	-500	901	406	
Δ Medicaid	-631	1240	562	

Notes: For each counterfactual in the columns (1)-(3), the table reports mean CWV for couples entering retirement as married, broken down by gender (wives and husbands), and for individuals who are initially single, also distinguished by gender. Δ Medicaid is the dollar change in per-household Medicaid spending relative to the baseline. The row "All (household-weighted)" reports the mean CWV evaluated at the household level. The row "All (individual-weighted)" reports the mean CWV evaluated at the individual level, weighting each married and single retiree equally.

variation (CWV), the initial non-housing wealth transfer that makes the economic agent indifferent to the counterfactual regime. Online Appendix F provides details on how we compute the CWV. Importantly, the collective model uncovers spouse-specific welfare effects using the spouse-specific value functions derived in Section 3, thereby revealing gender-differentiated policy consequences that would be masked under a unitary model relying solely on household-level CWV.[51]

For the counterfactuals, we vary the recipient of the homestead exemption used in Medicaid's means test, which is granted only to couples in the baseline.[52] To measure the fiscal impact, we compute the change in the present-discounted value of government expenditures over households' life cycles. As in Section 5.2, we construct initial conditions from the estimation sample that include both initially married and initially single households, with a household head aged 65–69. Column (4) in Table 12 reports the distribution of initial household types. Table G.5 in the Online Appendix reports the summary statistics by the initial household types. Throughout, our analysis is conducted in partial equilibrium, holding housing prices fixed.

First, we study the welfare impact when Medicaid no longer provides the homestead ex-

[51] Consistent with standard practice in the collective model literature, we treat the primitives of the model, including mapping $\lambda(I, z)$, as invariant to the Medicaid reforms we consider, which modify late-life budget constraints but not the long-run determinants of intrahousehold bargaining power.

[52] Our baseline follows a strict interpretation of Medicaid rules, under which couples, but not singles, qualify for the homestead exemption. Because enforcement varies across states, we interpret counterfactuals that extend the exemption and those that remove it as upper and lower bounds on welfare effects, respectively.

emption to couples in a nursing home, with welfare impacts reported in the first column of Table 12.[53] Households entering retirement as married are, on average, worse off by $658. By gender, wives lose about $800, while husbands' loss is $569. By allowing couples to remain homeowners longer, the homestead exemption slows dissaving and is particularly valuable for women, whose longer life expectancy strengthens their precautionary savings motives. Although men exhibit stronger bequest motives toward children, these motives are primarily relevant for high-PI households, who are unlikely to be affected by means-tested Medicaid. Table G.8 in the Online Appendix reports welfare impacts for initially married couples conditional on PI and gender. Consistent with this mechanism, the losses are concentrated among low-PI households, for whom means-tested Medicaid is most relevant.

Initially singles are not affected by the policy because they did not qualify for the exemption in the baseline. The household-level average welfare loss is $379. When we instead treat single individuals and each spouse within a couple as separate persons, the resulting individual-level average welfare loss is higher at $500. This difference is in part driven by wives' larger losses, which receive less weight in the household aggregation due to their lower Pareto weights on average. The savings in Medicaid spending amount to about $630 per household.

Next, we extend the homestead exemption to singles, such that all retirees' housing assets are disregarded by Medicaid's means test for nursing home use. Results are reported in Column (2) of Table 12. For those who enter retirement as married, extending the homestead exemption to periods of widowhood generates positive but very modest welfare gains. For initially singles, the welfare gains are substantial: $3,777 for women and $1,785 for men. The substantial welfare gains reflect the fact that individuals who enter retirement as single are poorer than those who enter as married, making means-tested Medicaid reforms more relevant for them. The larger gains for women are driven by their lower assets and incomes, which further increase the value of means-tested social insurance. It is also due to the fact that the homestead exemption allows them to retain homeownership for longer, which is particularly valuable given women's longer life expectancy and the resulting stronger incentives to save. Table G.7 in the Online Appendix reports the welfare impact for initially singles by PI and gender, showing that the effects are concentrated among low- and middle-PI households. The average welfare gain is about $1,410 per household, and the additional Medicaid spending is $1,240 per household.

Finally, in the third column of Table 12, we reform Medicaid so that the homestead ex-

[53]Table G.6 in the Online Appendix reports the corresponding changes in Medicaid recipiency conditional on current marital status and gender.

emption is granted to singles in a nursing home but not to couples. The impact on initially singles is the same as in the previous counterfactual, as they are newly granted the homestead exemption under both counterfactuals. For initially couples, the average welfare loss is nearly as large as when the homestead exemption is removed altogether, suggesting that retirees who enter retirement as married place a higher value on retaining the exemption early in retirement, while they are still married. As before, the welfare loss is larger for initially married women than for men. The welfare impact is positive on average ($1,032 at the household level and $406 per person), reflecting initially singles' greater valuation of the homestead exemption relative to couples. Initially singles are poorer on average and lack access to spousal care, leading to higher formal long-term care risks. This makes the homestead exemption, which preserves their housing assets while qualifying for public long-term care insurance, more valuable than it is for initially couples.

Taken together, these counterfactuals show that Medicaid's housing rules generate heterogeneous welfare consequences, with particularly strong effects for initial singles, for women, and for low-PI retirees. Importantly, our collective approach reveals meaningful within-household differences, with wives typically experiencing larger welfare changes due to their stronger precautionary savings motives. These gender disparities would be invisible under a unitary approach, underscoring the value of a collective model for evaluating policy reforms.

6 Conclusion

While many retirees spend a substantial portion of retirement married, little is known about intrahousehold bargaining in old age. By estimating a dynamic collective model of couples in retirement, we recover heterogeneity in preferences and bargaining power within married households. We use the estimated model to provide one of the first estimates of intrahousehold inequality in bargaining power and its implications for lifetime consumption, savings, and welfare in old age. We find that wives' mean lifetime consumption shares are close to parity, although there is substantial dispersion, with women who have higher income shares exerting greater influence in the household. Within couples, wives have stronger precautionary motives, reflecting longer life expectancy and a higher likelihood of requiring long-term care when no spouse is available to provide it. In contrast, men exhibit stronger bequest motives toward children, which are primarily relevant for high-PI households. As a result, reforms to welfare programs that encourage faster dissaving of retirement wealth disadvantage wives relative to husbands, since precautionary motives dominate and bequest motives play a limited role among households for which welfare programs are relevant. These findings reveal within-household distributional consequences that would be obscured under a unitary framework.

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Online Appendix (Not For Publication)

Intrahousehold Inequality and Savings in Retirement

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A Data and sample construction

This section provides details about the HRS sample, which is constructed from the biennial interview waves covering 1998-2014. We only consider individuals who were responsive and aged 65 and older in 1998; retired by 2000; and did not miss any interviews while alive. The main empirical sample consists of 6,475 respondents and 35,865 respondent-wave observations.

We measure housing assets as net value of primary residence.[1] We define homeowners as having strictly positive housing assets. We define renters as non-homeowners who are not residing in a nursing home. We measure non-housing assets as the sum of vehicles, businesses, IRA and Keogh accounts, stocks, mutual funds, investment trusts, checking, savings, money market accounts, CDs, bonds and T-bills. Our measure of income includes Social Security retirement income, pension, annuity, capital income, and other income.

The HRS asks respondents about their use of formal long-term care services including nursing homes. It also asks about the receipt of informal long-term care, defined as unpaid help associated with managing daily tasks. We observe the identity of informal caregivers, most of whom are either spouses or adult children.

We classify individuals as having long-term care needs if they report at least two limitations in activities of daily living (ADLs) or exhibit cognitive impairment. The availability of informal care provided by children is a dummy variable which is equal to one if a respondent says the number of children they believe will provide care when necessary exceeds zero.

Table G.1 reports the summary statistics of the sample used for descriptive analysis as well as structural estimation.

[1] Most retirees in our data have zero outstanding mortgages: the mean ratio of mortgage to housing value is 5%.

B Permanent income (PI) construction

We closely follow De Nardi et al. (2025) in specifying household income. We assume that log household income evolves according to:

$$\ln y_{it} = \kappa(t, f_{it}) + h(I_i) + \omega_{it}, \tag{1}$$

where $\kappa(t, f_{it})$ is a flexible function of age t and family structure f_{it} (i.e., couple, single man, or single woman), and ω_{it} captures measurement error. The term I_i denotes the individual's percentile rank in the PI distribution, which is fixed over time. We begin by estimating a fixed effects model:

$$\ln y_{it} = \kappa(t, f_{it}) + \alpha_i + \omega_{it}. \tag{2}$$

We then compute I_i as the percentile rank of the individual's sample mean of $\ln y_{it} - \hat{\kappa}(t, f_{it})$, using $\hat{\kappa}(t, f_{it})$ from the fixed effects estimates. Next, we estimate $h(I_i)$ by fitting:

$$\ln y_{it} - \kappa(t, f_{it}) = h(I_i) + \omega_{it}, \tag{3}$$

where $h(I_i)$ is specified as a fifth-order polynomial in the estimated PI percentile. The function $\kappa(t, f_{it})$ is modeled as a third-order polynomial in age, with family structure dummies and interactions with age trends. The predicted income is then given by:[2]

$$\ln \hat{y}_{it} = \hat{\kappa}(t, f_{it}) + \hat{h}(\hat{I}_i). \tag{4}$$

As in De Nardi et al. (2025), the index I_i does not vary with marital transition.

For our specification of Pareto weights, we also need to define the wife's share of the household income for couples. To relate individual income components to the household-level PI measure constructed above, we estimate the following household-level regression:

$$\hat{h}(\hat{I}_i) = \beta_0 + \beta_1 \text{income}_{\text{husband},i} + \beta_2 \text{income}_{\text{wife},i} + \epsilon_i, \tag{5}$$

where $\text{income}_{g,i}$ denotes the average income of household member $g \in \{\text{husband}, \text{wife}\}$ over the sample period.[3] Based on the estimated coefficients, we define the wife's income share

[2] Based on this approach, a regression of $\ln y_{it}$ on $\ln \hat{y}_{it}$ gives an R^2 of 0.728. Also, we find no evidence of positive serial correlation of the residuals from this regression (across any two waves, serial correlation is -0.034).

[3] In the HRS, incomes are reported at the individual level except for capital and other income which are reported at the household level. To define each spouse's individual income, we divide these two measures by

as:

$$\frac{\hat{\beta}_2 \cdot \text{income}_{\text{wife},i}}{\hat{\beta}_1 \cdot \text{income}_{\text{husband},i} + \hat{\beta}_2 \cdot \text{income}_{\text{wife},i}}. \tag{6}$$

The mean of the resulting income share across households is 0.34.

C Homeownership around spousal death

We conduct an event study analysis to examine changes in the homeownership rate around spousal death. For this exercise, the sample consists of married individuals who were homeowners four years prior to their spouse's death and who remain alive four years afterward. From this sample, we estimate

$$Y_{it} = \sum_{k=-4}^{4} \beta_k I_{itk} + \gamma X_{it} + \tau_t + \varepsilon_{it}, \tag{7}$$

where the dependent variable is one if individual i is still a homeowner and zero otherwise. I_{itk} represents an indicator for being k years since the spousal death. The vector X_{it} includes gender, birth cohort, an indicator for having a child, long-term care needs, income, and non-housing assets. τ_t represents year fixed effects. Figure G.4 reports the estimates of β_k. The results show that there is a statistically significant reduction in homeownership rates following spousal death.

Table G.2 uses the same sample as in the event study and reports living arrangements four years after spousal death. Among those that liquidate their house and become non-homeowners, 67% move into rented units, 25% move into a nursing home, and 8% move into their children's home.

D Model appendix

D.1 Incorporation of side bequests and its challenges

In this appendix, we outline how side bequests could be incorporated into our model and explain why the available data do not allow credible identification without strong assumptions.

two.

When a deceased individual is survived by a spouse, let us assume that a fraction of the bequest is allocated to children, following De Nardi et al. (2025). The crucial difference is that in their unitary framework, the surviving spouse determines the size of the side bequest. Specifically, the surviving spouse chooses how much to allocate to children to maximize the sum of the surviving spouse's utility from the side bequest and continuation value as a single retiree. In contrast, our collective framework requires us to explicitly specify how both the deceased and the surviving spouses value side bequests. The amount of side bequests will be set while both spouses are alive, as the outcome of a collective decision-making process. Under our full commitment framework, we assume that these predetermined side bequests are enforceable, effectively treating them as if specified in a will.

We can model the bequest preferences of a deceased individual who is survived by a spouse as

$$v^{C,g}(x_t) = v^{\mathrm{dec},g}(\tau \cdot b_t) + \alpha V^{S,g'}((1-\tau) \cdot b_t), \tag{8}$$

where b_t is the remaining household assets at the first spouse's death, and τ is the predetermined share of the bequest allocated to children. The function $v^{\mathrm{dec},g}$ represents the deceased individual's utility from leaving bequests to children when there is a surviving spouse. While these preferences may coincide with $v^{S,g}$, the utility from leaving bequests to children as a single individual, this is not obvious. As in the main text, we allow the deceased to care for the surviving spouse, captured via the pure altruism parameter, α. The surviving spouse's value depends on elements other than remaining assets after side bequests, which we suppress for notational simplicity.[4]

For the collective framework, we also need to specify how the surviving spouse values side bequests. For instance, if the husband dies first and a predetermined fraction τ of remaining wealth goes to children, the surviving wife may derive some utility from this transfer. From her perspective, however, this transfer is not a real bequest as she is still alive, so it more closely resembles an inter-vivos transfer. Therefore, we do not model the surviving spouse's utility as $v^{S,g}(\cdot)$, which reflects bequest utility upon their own death. Nor is it obvious that the surviving spouse's valuation of side bequests is identical to that of their deceased spouse, $v^{\mathrm{dec},g}$. We therefore denote the surviving spouse's preferences for side bequests in

[4]While it is plausible that $v^{\mathrm{dec},g}$ depends only on total assets, the surviving spouse's continuation value $V^{S,g'}$ depends on the full state vector, x_t, which includes liquid and housing assets as separate state variables. A full incorporation of side bequests would therefore require modeling how side bequests are funded, including the share drawn from liquid versus housing assets. We abstract from this additional layer because our point is to demonstrate that preferences over side bequests are already difficult to identify even before accounting for asset composition.

this context as $v^{\text{surv},g}$.

It is not feasible to identify the altruism parameter toward the spouse, α, alongside both $v^{\text{dec},g}$ and $v^{\text{surv},g}$ given the data we have. Even under the simplifying assumption that preferences for leaving bequests to children at one's own death are the same regardless of whether there is a surviving spouse, i.e., $v^{\text{dec},g} = v^{S,g}$, identification of α and $v^{\text{surv},g}$ remains infeasible. That is, we would not be able to tell how much of the side bequests is coming from the deceased's altruism for the surviving spouse vs. the surviving spouse's preference for side bequests (or inter-vivos transfers from her perspective).

If we impose a further assumption that the surviving spouse's preferences for side bequests are the same as her preferences for terminal bequests at her own death, $v^{\text{surv},g} = v^{S,g}$, then identification becomes feasible as discussed in Section 4.2.1. Specifically, we can use the evolution of singles' savings to obtain information about $v^{S,g}$ and the evolution of couples' savings as particularly useful variations to identify altruism toward the spouse. However, this is a strong assumption that is unlikely to hold in reality. For example, De Nardi et al. (2025) find that in their unitary framework, individuals' preferences for terminal bequests are stronger than those for side bequests. We therefore choose to abstract from side bequests. Instead, we focus on bequests left to the surviving spouse, which dominate empirically (85.6% of the estate) and allow us to study altruism toward the surviving spouse, a component only identifiable in a non-unitary framework.

D.2 Recursive formulation for singles' problem

We provide a recursive formulation for a single household's problem. In each period, a single household's state vector is given as

$$x_t = (a_t, \tilde{h}_{t-1}, s_t^g; I, ic_{child}), \tag{9}$$

where a_t is the non-housing wealth, $\tilde{h}_{t-1}$ is the housing wealth, and s_t^g is the health status of single gender $g \in \{W, M\}$. Time-invariant state variables are PI (I) and the availability of informal care from children (ic_{child}).

The household's choice vector is $q_t = (\hat{c}_t, D_t, R_t)$ where $\hat{c}_t \geq 0$ is non-housing consumption, $D_t \in \{0, 1\}$ represents the house selling choice, and $R_t \geq 0$ is the rent choice.

Denote the survival probability of gender g in period t by p_t^g which also depends on current health, PI, and marital status. A recursive formulation for a single household's problem for gender g is

$$V_t^{S,g}(x_t) = \max_{q_t} \quad u^g(c_t, h_t) + \beta p_t^g E[V_{t+1}^{S,g}(x_{t+1})|x_t, q_t] + \beta(1-p_t^g)v^{S,g}(b_{t+1}) \tag{10}$$

subject to budget constraints. $V_t^{S,g}$ denotes the value function of a single retiree of gender g in t. If they survive, they continue the next period again as a single retiree, where the expectation operator is taken with respect to health statuses of the next period. If they die, their remaining assets are left as bequests to children or other heirs, in which case they derive bequest utility of $v^{S,g}(b_{t+1})$. The terminal value function is given as $V_T^{S,g}(x_T) = v^{S,g}(b_T)$.

E Estimation procedure

We implement indirect inference by following the limited information Bayesian approach as in Fernandez-Villaverde et al. (2016). Let Θ denote the parameters of interest and $\hat{\mathbf{m}}$ denote the vector of M parameters of auxiliary models from the empirical data. The objective function of $\hat{\mathbf{m}}$ conditional on Θ is

$$f(\hat{\mathbf{m}}|\Theta) = -\left(\hat{\mathbf{m}} - \mathbf{m}(\Theta)\right)' G \left(\hat{\mathbf{m}} - \mathbf{m}(\Theta)\right), \tag{11}$$

where $\mathbf{m}(\Theta)$ is the parameters of auxiliary models based on the simulated data given Θ and G is a weighting matrix. Bayes' theorem tells us that the posterior density $f(\Theta|\hat{\mathbf{m}})$ is proportional to the product of $f(\hat{\mathbf{m}}|\Theta)$ and the prior $p(\Theta)$, which we take to be uniform[5]:

$$f(\Theta|\hat{\mathbf{m}}) \propto f(\hat{\mathbf{m}}|\Theta)p(\Theta). \tag{12}$$

We use a multiple-block random-walk Metropolis–Hastings (RWMH) algorithm to obtain a sequence of random samples from the posterior distribution. As discussed in Section 4.2.1, different subsets of parameters are primarily informed by different sets of empirical targets. Accordingly, we organize the blocks of the algorithm so that each parameter block is aligned with the auxiliary moments to which it is most sensitive.

We denote the objects from auxiliary models to match as $\hat{\mathbf{m}} = [\hat{\mathbf{m}}_1, \hat{\mathbf{m}}_2]$. $\hat{\mathbf{m}}_1$ contains the targets reported in Tables 8 and 9, and $\hat{\mathbf{m}}_2$ includes the wealth moments reported in Figure 2. Accordingly, we partition the parameter vector into two blocks: one block (block 1) contains

[5] In cases where prior knowledge or beliefs regarding parameters are available, informative priors can be employed to integrate them into Bayesian estimation. The selection of suitable informative prior distributions remains a subject of debate, representing an avenue for future research.

the parameters that are central to governing the regression coefficients and the rate moments in $\hat{\mathbf{m}}_1$, and the other block (block 2) contains the parameters that are central to governing wealth moments in $\hat{\mathbf{m}}_2$. Specifically, we write the parameter space as $\Theta = (\Theta^1, \Theta^2)$, where $\Theta^1 = (\beta_I, \beta_z, \beta_0, \psi^W, \psi^M, \bar{a}_{nh=0})$ and Θ^2 collects the remaining model parameters, including those governing bequest motives.

Starting from an initial value $\Theta_0 = (\Theta_0^1, \Theta_0^2)$, the algorithm works as follows. For iteration $j = 1, \ldots, M$ and for block $k = 1, 2$:

- Propose a value $\tilde{\Theta}^k$ for the kth block, conditional on Θ_{j-1}^k for the kth block and the current value of the other block (Θ^{-k}). Θ^{-k} stands for the remaining block except for the kth block.[6]
- Compute the acceptance probability $\alpha^k = \min\left\{1, \frac{f(\tilde{\Theta}^k|\Theta^{-k}, \hat{\mathbf{m}})}{f(\Theta_{j-1}^k|\Theta^{-k}, \hat{\mathbf{m}})}\right\}$. Update the kth block as

$$\Theta_j^k = \begin{cases} \tilde{\Theta}^k & \text{w.p. } \alpha^k \\ \Theta_{j-1}^k & \text{w.p. } (1 - \alpha^k) \end{cases}$$

For each iteration, we first update block 1's parameters conditional on the previous iteration's value for the blocks 1 and 2. Then we sequentially update the remaining model parameter block conditional on the updated block 1.[7]

We apply the multiple-block RWMH algorithm to simulate draws from the posterior $f(\Theta|\hat{\mathbf{m}})$ with uniform priors. The posterior distribution is characterized by a sequence of 4000 draws.

F Welfare measure

Our measure of welfare is the compensating wealth variation (CWV), the initial non-housing wealth transfer that makes the economic agent indifferent to the counterfactual regime. For notational simplicity, in this appendix, we write value functions as functions of liquid assets only, implicitly holding the remaining state variables in the full state vector x_t fixed.

[6] In our application, $\Theta^{-1} = \Theta^2$ and $\Theta^{-2} = \Theta^1$.

[7] For block 1, we begin with a uniform weighting matrix G, but increase the weights on the regression coefficients relative to the other rate moments, reflecting our priority to match these coefficients more closely as they carry the key information in identifying Pareto weights. Similar weighting strategies have been used in the literature that emphasize putting additional weights on objects that are key in identifying economically important parameters (Sauer and Taber, 2021). For block 2, we use a diagonal weighting matrix based on the inverse of the estimated variance–covariance matrix of the dollar-valued moments.

Singles' CWV is denoted by $\Delta^{S,g}$, defined as

$$V_t^{S,g}(a_t + \Delta^{S,g}) = \tilde{V}_t^{S,g}(a_t) \tag{13}$$

for $t = 1$ and gender $g \in \{W, M\}$, where $\tilde{V}^{S,g}$ is the value under the counterfactual.

Couples' household-level CWV is denoted by Δ^C, defined as

$$V_t^C(a_t + \Delta^C) = \tilde{V}_t^C(a_t) \tag{14}$$

for $t = 1$, where $\tilde{V}^C$ is the household's value under the counterfactual.

To analyze intrahousehold effects, we compute analogous measures for each spouse $g \in \{W, M\}$, with $\Delta^{C,g}$ defined by

$$V_t^{C,g}(a_t + \Delta^{C,g}) = \tilde{V}_t^{C,g}(a_t). \tag{15}$$

Because the household's value is an aggregation of the spouses' values, these CWVs are linked by

$$V_t^C(a_t + \Delta^C) = \lambda V_t^{C,W}(a_t + \Delta^{C,W}) + (1 - \lambda) V_t^{C,M}(a_t + \Delta^{C,M}). \tag{16}$$

Linearizing around baseline wealth yields an intuitive relationship:

$$\Delta^C = \frac{\lambda \Delta^W \frac{\partial V^{C,W}}{\partial a} + (1 - \lambda) \Delta^M \frac{\partial V^{C,M}}{\partial a}}{\lambda \frac{\partial V^{C,W}}{\partial a} + (1 - \lambda) \frac{\partial V^{C,M}}{\partial a}}. \tag{17}$$

In words, the couple's compensating wealth equivalent is a weighted average of the wife's and husband's, where the weights reflect each spouse's marginal value of wealth. Using (15) and (16), one can easily show that $\min(\Delta^W, \Delta^M) \leq \Delta^C \leq \max(\Delta^W, \Delta^M)$.

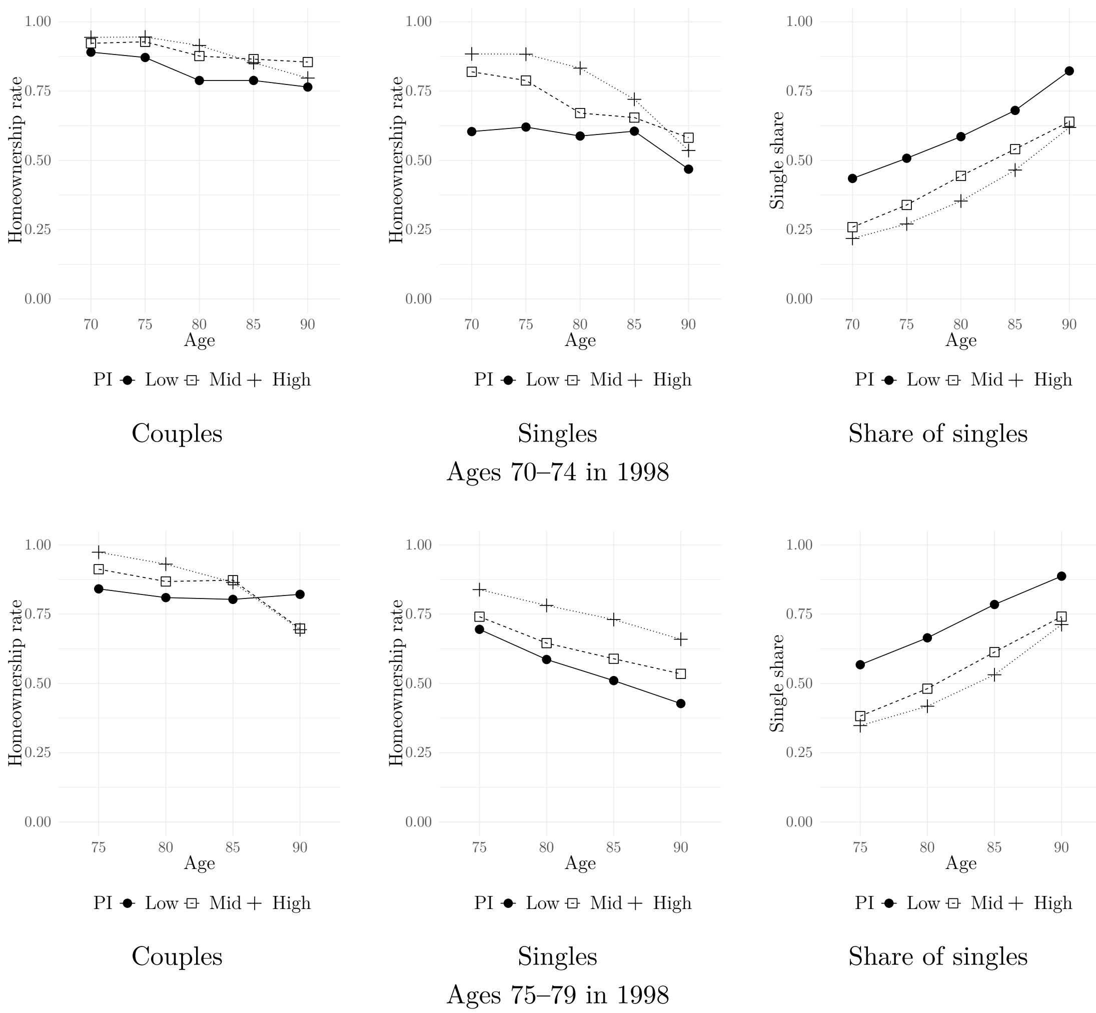
G Additional tables and figures

Table G.1: Summary statistics of the HRS sample

	(1) Single			(2) Married		
	Mean	Median	Std.	Mean	Median	Std.
Age	80.53	81.00	6.68	77.41	77.00	5.90
Female	0.79	1.00	0.41	0.50	0.00	0.50
Housing assets	101606	44700	202082	187710	136000	256124
Non-housing assets	203415	23430	762834	502049	163155	1161399
Homeowner	0.59	1.00	0.49	0.88	1.00	0.33
Renter	0.31	0.00	0.46	0.11	0.00	0.31
Have LTC needs	0.24	0.00	0.43	0.13	0.00	0.33
Household income	31848	20444	63015	64367	45552	81832
Wife's income share	.	.	.	0.31	0.30	0.13
Believe children will provide LTC	0.74	1.00	0.44	0.76	1.00	0.43
Observations	18895			16970		

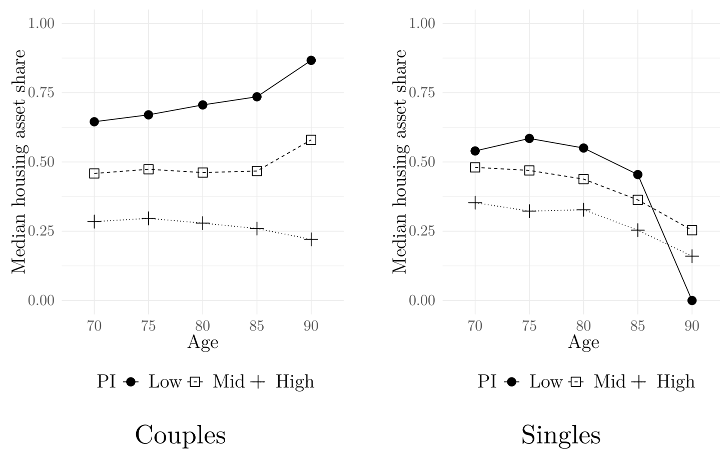
Notes: The HRS sample (1998–2014) used which consists of retired individuals aged 65 and older in 1998. The table reports the summary statistics conditional on current marital status. Observations are at the individual–wave level.

Figure G.1: Homeownership rates



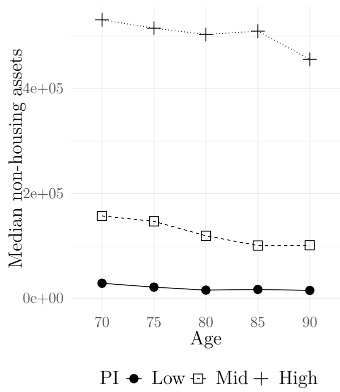
Notes: The figure reports homeownership rates by current marital status, PI, age, and birth cohort, focusing on those aged 70–74 and 75–79 in 1998.

Figure G.2: Median housing asset share

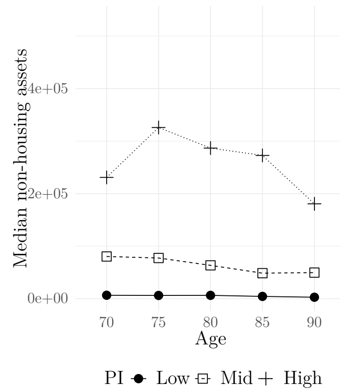


Notes: The figure shows the median housing asset share by current marital status, PI terciles, and age group. Housing asset share is defined as the ratio of housing assets to total assets. The third plot in Figure 1 reports the share of currently singles conditional on PI and age group.

Figure G.3: Median non-housing assets

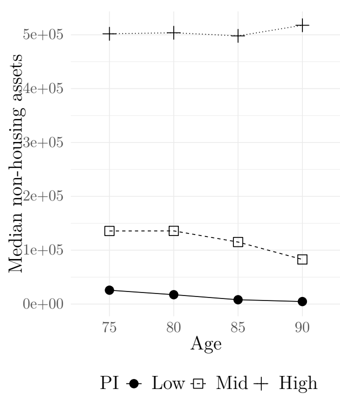


Couples

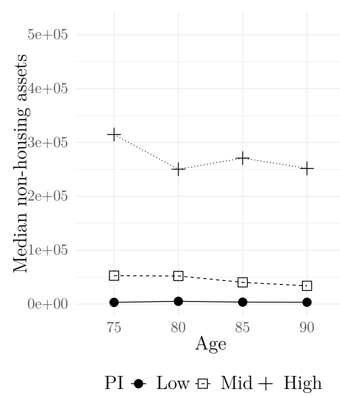


Singles

Ages 70–74 in 1998



Couples

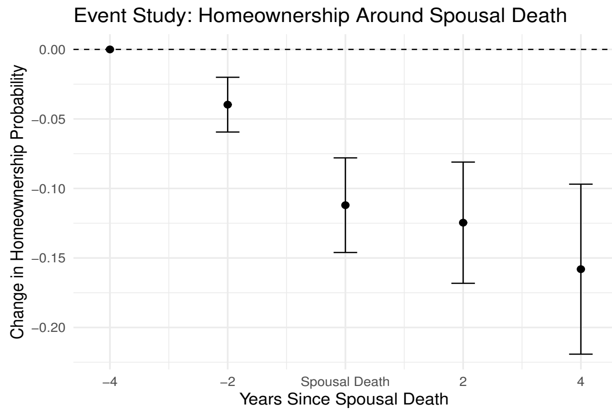


Singles

Ages 75–79 in 1998

Notes: The figure shows the median non-housing assets ($) by birth cohort, current marital status, PI terciles, and age group. Figure G.1 reports the share of singles for each birth cohort, conditional on PI and age group.

Figure G.4: Homeownership around spousal death



Notes: The sample consists of individuals who were homeowners four years prior to their spouse's death and who remain alive four years afterward. The figure reports the estimates of β_k in equation (7) of the Online Appendix C along with their 95% confidence interval. Standard errors are clustered at the household level.

Table G.2: Living arrangements after spousal death

	Non homeowners	Still homeowners
Move into nursing home	0.248	0.080
Move into kids' home	0.083	0.017
Move into rented units	0.669	–
Observations	109	349

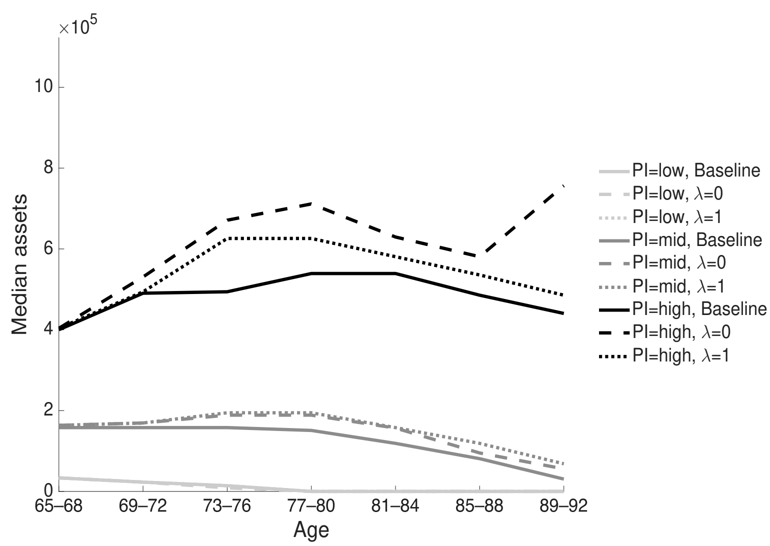
Notes: The table reports living arrangements four years after spousal death for the sample used in the event study analysis reported in Figure G.4.

Table G.3: Wife's income share and household consumption-savings

	(1) Consumption rate		(2) Log of consumption		(3) Savings rate		(4) Change in assets	
Wife's income share	-0.34**	(0.16)	-0.29***	(0.11)	1.22**	(0.59)	78361.31***	(28889.56)
Husband age	-0.00	(0.01)	-0.01	(0.00)	-0.03**	(0.02)	-2031.79**	(856.35)
Wife age	-0.02***	(0.01)	-0.02***	(0.00)	-0.01	(0.02)	-231.59	(834.54)
Housing assets from prev. wave (in 100K)	0.18***	(0.02)	0.14***	(0.01)	-0.27***	(0.07)	-5650.64	(3480.55)
Liquid assets from prev. wave (in 100K)	0.02***	(0.01)	0.01***	(0.00)	-0.56***	(0.04)	-33427.40***	(2149.12)
Household income (in 100K)	0.18**	(0.07)	0.15***	(0.05)	3.65***	(0.43)	201698.95***	(24653.15)
PI (low)	0.00	(.)	0.00	(.)	0.00	(.)	0.00	(.)
PI (mid)	-0.79***	(0.10)	0.26***	(0.04)	0.43**	(0.19)	10315.34	(7445.09)
PI (high)	-1.75***	(0.11)	0.30***	(0.05)	1.32***	(0.28)	45659.16***	(14154.88)
Husband has LTC needs	-0.10	(0.06)	-0.06	(0.05)	-0.19	(0.20)	-5539.55	(9802.13)
Wife has LTC needs	-0.01	(0.08)	-0.03	(0.05)	-0.58**	(0.24)	-24742.61**	(10685.84)
Number of children	-0.02**	(0.01)	-0.01	(0.01)	-0.00	(0.03)	-903.73	(1483.98)
Husband's education years	0.02**	(0.01)	0.01***	(0.00)	0.03	(0.02)	2713.43**	(1156.69)
Wife's education years	0.02	(0.01)	0.02**	(0.01)	-0.07*	(0.03)	-1754.63	(1482.48)
Out-of-pocket medical spending	0.00**	(0.00)	0.00**	(0.00)	-0.00*	(0.00)	-0.47**	(0.19)
Husband has Medicare	-0.08	(0.24)	0.06	(0.08)	-0.88*	(0.49)	-38481.05	(31313.37)
Wife has Medicare	-0.16	(0.25)	0.10	(0.06)	1.00*	(0.55)	64151.98**	(30467.53)
Husband has private LTC insurance	0.12	(0.09)	0.09*	(0.06)	0.74*	(0.39)	42022.41**	(17736.39)
Wife has private LTC insurance	-0.07	(0.08)	-0.03	(0.05)	-0.50	(0.32)	-29637.50*	(15724.12)
Observations	1703.00		1703.00		11012.00		11012.00	
Adjusted R^2	0.40		0.42		0.12		0.18	
Mean of dependent variable	1.39		10.87		-0.27		-13749.23	

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the household level (in parentheses). The table reports coefficient estimates not included in Table 1 of the main text. Not all controls are shown, including birth cohort fixed effects.

Figure G.5: Counterfactual wealth trajectories for currently singles



Notes: The figure plots median wealth among currently singles, defined as the sum of liquid and housing assets, by PI and age, with darker colors indicating higher PI. Solid lines show the baseline; dashed lines show the counterfactual in which the husband's Pareto weight is one ($\lambda = 0$); and dotted lines show the counterfactual in which the wife's Pareto weight is one ($\lambda = 1$).

Table G.4: Gender and consumption-savings among singles

	(1) Consumption rate		(2) Consumption rate		(3) Savings rate		(4) Savings rate	
Female	-0.14***	(0.04)	-0.26**	(0.11)	0.17	(0.11)	0.28	(0.18)
Age	-0.02***	(0.00)	-0.02***	(0.00)	-0.01	(0.01)	-0.01	(0.01)
Housing assets from prev. wave (in 100K)	0.15***	(0.02)	0.15***	(0.02)	-0.40***	(0.06)	-0.40***	(0.05)
Liquid assets from prev. wave (in 100K)	0.02***	(0.00)	0.02***	(0.00)	-0.72***	(0.04)	-0.72***	(0.04)
Household income (in 100K)	0.12**	(0.05)	0.14***	(0.05)	3.81***	(0.33)	3.74***	(0.33)
PI (low)	0.00	(.)	0.00	(.)	0.00	(.)	0.00	(.)
PI (mid)	-0.62***	(0.05)	-0.79***	(0.11)	0.35***	(0.11)	0.20	(0.24)
PI (high)	-1.27***	(0.06)	-1.41***	(0.12)	1.83***	(0.18)	2.41***	(0.36)
Have LTC needs	-0.07	(0.05)	-0.06	(0.05)	-0.44***	(0.11)	-0.45***	(0.11)
Number of children	0.01	(0.01)	0.01	(0.01)	-0.04***	(0.02)	-0.05***	(0.02)
Education years	0.03***	(0.01)	0.03***	(0.01)	-0.00	(0.01)	-0.01	(0.01)
Out-of-pocket medical spending	0.00**	(0.00)	0.00**	(0.00)	-0.00	(0.00)	-0.00	(0.00)
Have Medicare	0.04	(0.09)	0.03	(0.09)	-0.31	(0.53)	-0.30	(0.53)
Have private LTC insurance	0.00	(0.03)	-0.00	(0.03)	0.22	(0.16)	0.25	(0.16)
PI (low) × Female			0.00	(.)			0.00	(.)
PI (mid) × Female			0.20*	(0.12)			0.20	(0.24)
PI (high) × Female			0.18	(0.12)			-0.68**	(0.33)
Observations	3537.00		3537.00		13711.00		13711.00	
Adjusted R^2	0.26		0.26		0.14		0.14	
Mean dep. var	1.08		1.08		-0.41		-0.41	

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the household level (in parentheses). The table reports coefficient estimates not included in Table 3 of the main text. Sample consists of single respondents from HRS 1998-2014. Consumption questions were asked only to a randomly selected subsample. Not all controls are shown, including birth cohort fixed effects.

Table G.5: Summary statistics by initial household types

Initial househole type:	(1) Couples		(2) Single females		(3) Single males	
	Mean	Std.	Mean	Std.	Mean	Std.
Non-housing assets	602824.25	1623007.80	123272.89	311352.20	259931.47	925203.57
Housing assets	173090.44	206789.73	87581.36	121143.64	63352.18	154457.30
Homeowner	0.93	0.26	0.68	0.47	0.48	0.50
Renter	0.07	0.26	0.30	0.46	0.49	0.50
Have LTC needs	0.06	0.24	0.12	0.33	0.12	0.33
Household income	76453.44	100409.12	30244.54	40198.26	50873.95	121976.16

Notes: The sample consists of initial conditions for the counterfactuals in Sections 5.2 and 5.3, where the household head is aged 65–69.

Table G.6: Percent changes (%) in Medicaid recipiency rate

	(1)	(2)	(3)
Homestead exemption:	Nobody	Everybody	Singles
Married	-4.1	-0.3	-4.0
Single females	1.6	0.6	2.3
Single males	0.9	3.4	4.1

Notes: The table reports percent changes from the baseline Medicaid recipiency rate, conditional on current marital status and gender.

Table G.7: Welfare effects of homestead exemption for initially singles

	Mean welfare effect
Low PI	W: 6029 M: 2108
Middle PI	W: 1208 M: 2138
High PI	W: 396 M: 0

Notes: Mean compensating wealth equivalents for initially single women (W) and men (M) when singles are granted the homestead exemption.

Table G.8: Intrahousehold welfare effects of counterfactual Medicaid for initially couples

Homestead exemption:	Nobody	Everybody	Singles
Low PI	C: -2388 W: -2243 M: -2689	C: 87 W: 108 M: 25	C: -2284 W: -2126 M: -2557
Middle PI	C: -231 W: -713 M: 209	C: 47 W: 90 M: -24	C: -191 W: -782 M: 307
High PI	C: 0 W: 0 M: 0	C: 0 W: 0 M: 0	C: 0 W: 0 M: 0

Notes: Mean compensating wealth equivalents (CWV) by PI group for initially married couples (C), reported separately for wives (W) and husbands (M).

References

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